CSC3002F - Computer Networks

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Search

MYBROADBAND
TRUSTED IN TECH

Big Internet problems in South Africa

By Hanno Labuschagne



Several Internet users in South Africa have reported connectivity issues due to an IP transit problem on a major long-distance fibre infrastructure operator.

Online service outage tracking platform Downdetector showed a big increase in reports indicating problems on Seacom from around 10:30 on Tuesday, 4 November 2025.

Around the same time, several international online services — including Discord, OpenAI, and Cloudflare — were also seeing a large number of reports.

MyBroadband Forum members also complained of slow or no access to international websites and services, including YouTube and Twitch.

INTERNET DOWN

Teraco and Seacom cable fault brings about digital load shedding for some

Another internet outage in South Africa shows us that the internet, much like an ogre and onion analogy, has layers. And you'd better have redundancy.



A Teraco and Seacom outage took down critical South African internet service on Tuesday. (Photo: iStock)

DIVE DEEPER | SPEED READ | SUMMARY | LISTEN

By Lindsey Schutters 4 Nov 2025 0

It was a cable break, but not in the way you'd expect. Service provider notifications brought it to the attention of Daily Maverick's news desk, and then some excellent Downdetector sleuthing by MyBroadBand confirmed it.

hypertext Business Lifestyle Energy Education Features Security Reviews Competitions Parti



NEWS

FACEBOOK X LINKEDIN WHATSAPP REDDIT

Huge AWS outage knocks online services in South Africa

Luis Monzon 20th October 2025

- **In what looks like a major global outage, an issue with Amazon Web Services (AWS) is affecting how South Africans use the internet.**
- **AWS says it is exploring multiple methods to resolve the issue and may have already identified the problem.**
- **It is yet another example of how the internet is hinged on a small number of corporations.**

Business — 18 Nov 2025

Systems returning to normal after global Cloudflare outage hits internet users

By Newsday



A massive global outage at web-infrastructure giant Cloudflare caused widespread disruption to major online platforms on Tuesday.

It knocked services such as X, ChatGPT, and numerous South African websites offline for hours before the company began restoring traffic late in the afternoon.

Cloudflare, whose network carries roughly a fifth of global web traffic and shields websites from traffic surges and cyberattacks,first detected "internal service degradation" at around 13:30 SAST.

Introduction

➢ how does 'information' flow from Google.com to your device?

- use *ping* to measure 'delay' or 'latency'

```
UCTAdmins-MBP:~ josiahchavula$ ping www.google.com
PING www.google.com (64.233.165.147): 56 data bytes
64 bytes from 64.233.165.147: icmp_seq=0 ttl=58 time=278.287 ms
64 bytes from 64.233.165.147: icmp_seq=1 ttl=58 time=219.355 ms
64 bytes from 64.233.165.147: icmp_seq=2 ttl=58 time=325.308 ms
64 bytes from 64.233.165.147: icmp_seq=3 ttl=58 time=244.454 ms
64 bytes from 64.233.165.147: icmp_seq=4 ttl=58 time=262.087 ms
64 bytes from 64.233.165.147: icmp_seq=5 ttl=58 time=285.870 ms
64 bytes from 64.233.165.147: icmp_seq=6 ttl=58 time=304.718 ms
64 bytes from 64.233.165.147: icmp_seq=7 ttl=58 time=222.744 ms
64 bytes from 64.233.165.147: icmp_seq=8 ttl=58 time=243.122 ms
64 bytes from 64.233.165.147: icmp_seq=9 ttl=58 time=266.922 ms
64 bytes from 64.233.165.147: icmp_seq=10 ttl=58 time=222.005 ms
64 bytes from 64.233.165.147: icmp_seq=11 ttl=58 time=216.157 ms
64 bytes from 64.233.165.147: icmp_seq=12 ttl=58 time=296.265 ms
64 bytes from 64.233.165.147: icmp_seq=13 ttl=58 time=249.414 ms
64 bytes from 64.233.165.147: icmp_seq=14 ttl=58 time=219.183 ms
```

~**250ms** from Afrihost CPT to Google.com

Afrihost resolves Google.com to IP 64.233.165.147 (a server in Europe)
while
UCT resolves Google.com to IP 216.58.223.132 (a server in Johannesburg)

```
jchavula@jchavula-1:~$ ping www.google.com
PING www.google.com (216.58.223.132) 56(84) bytes of data.
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=1 ttl=111 time=17.4 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=2 ttl=111 time=17.2 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=3 ttl=111 time=17.2 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=4 ttl=111 time=17.1 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=5 ttl=111 time=17.4 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=6 ttl=111 time=17.3 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=7 ttl=111 time=17.3 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=8 ttl=111 time=17.4 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=9 ttl=111 time=17.2 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=10 ttl=111 time=17.2 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=11 ttl=111 time=17.2 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=12 ttl=111 time=17.1 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=13 ttl=111 time=17.0 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=14 ttl=111 time=17.1 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=15 ttl=111 time=16.9 ms
64 bytes from jnb02s01-in-f4.1e100.net (216.58.223.132): icmp_seq=16 ttl=111 time=17.1 ms
^C
--- www.google.com ping statistics ---
16 packets transmitted, 16 received, 0% packet loss, time 15020ms
rtt min/avg/max/mdev = 16.986/17.254/17.478/0.219 ms
```

~**17ms** from UCT network to Google.com

Introduction

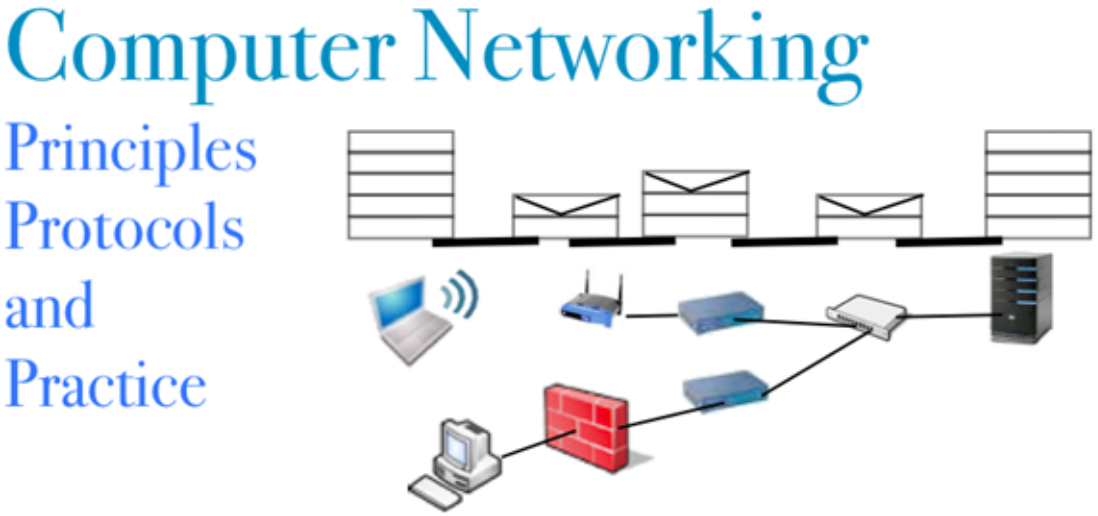
- how does ‘information’ flow from Google.com to your device?
  - use *traceroute* (tracert on Windows) to see the network path

```
jchavula@jchavula-1:~$ traceroute -A www.google.com
traceroute to www.google.com (216.58.223.132), 30 hops max, 60 byte packets
 1  _gateway (137.158.61.129) [AS36982]  0.537 ms  0.807 ms  1.069 ms
 2  10.1.23.42 (10.1.23.42) [*]  0.802 ms 10.1.23.34 (10.1.23.34) [*]  0.883 ms 10.1.23.42 (10.1.23.42) [*]  0.975 ms
 3  10.1.23.33 (10.1.23.33) [*]  1.226 ms 10.1.23.41 (10.1.23.41) [*]  0.880 ms 10.1.23.33 (10.1.23.33) [*]  1.360 ms
 4  vlan421.uc-edge.net.uct.ac.za (137.158.248.17) [AS36982]  1.558 ms  1.254 ms  1.658 ms
 5  154.114.124.0 (154.114.124.0) [AS2018]  1.456 ms  1.461 ms  1.473 ms
 6  et-1-1-5-0-cpt3-pe1.net.tenet.ac.za (155.232.64.71) [AS2018]  2.142 ms  2.067 ms  2.020 ms
 7  et-0-1-0-0-isd1-pe1.net.tenet.ac.za (155.232.1.149) [AS2018]  18.563 ms  18.625 ms  18.496 ms
 8  lt-1-1-0-1-isd1-pe1.net.tenet.ac.za (155.232.128.76) [AS2018]  18.668 ms lt-0-1-0-1-isd1-pe1.net.tenet.ac.za (155.232.128.70) [AS2018]  18.289 ms  18.235 ms
 9  72.14.211.150 (72.14.211.150) [AS15169]  18.182 ms  18.089 ms  18.093 ms
10  74.125.245.209 (74.125.245.209) [AS15169]  19.060 ms 74.125.245.193 (74.125.245.193) [AS15169]  17.996 ms *
11  172.253.66.0 (172.253.66.0) [AS15169]  17.917 ms jnb02s01-in-f4.1e100.net (216.58.223.132) [AS15169/AS19281]  17.861 ms 172.253.65.255 (172.253.65.255) [AS15169]  20.073 ms
```

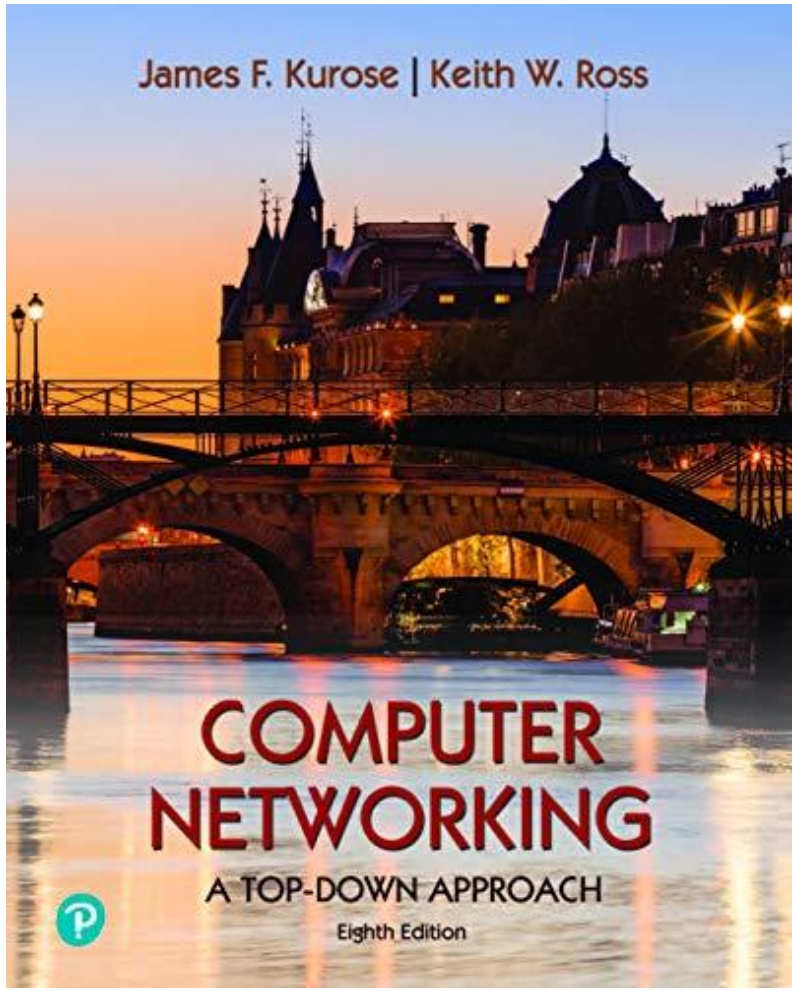
```
UCTAdmins-MBP:~ josiahchavula$ traceroute -a www.google.com
traceroute: Warning: www.google.com has multiple addresses; using 173.194.220.147
traceroute to www.google.com (173.194.220.147), 64 hops max, 52 byte packets
 1  [AS0] dlinkrouter.dlink (192.168.0.1)  3.027 ms  3.561 ms  3.063 ms
 2  [AS37611] 165-73-99-129.ip.afrihost.capetown (165.73.99.129)  4.485 ms  3.519 ms  6.790 ms
 3  [AS37611] 169-1-21-147.ip.afrihost.co.za (169.1.21.147)  4.441 ms  3.270 ms  3.900 ms
 4  [AS37611] 169-1-21-131.ip.afrihost.co.za (169.1.21.131)  5.291 ms  3.808 ms  5.536 ms
 5  [AS0] 100.127.2.174 (100.127.2.174)  22.656 ms  23.442 ms  25.399 ms
 6  [AS15169] 72.14.216.130 (72.14.216.130)  24.420 ms  24.178 ms  23.441 ms
 7  [AS0] 10.252.254.94 (10.252.254.94)  29.372 ms
    [AS0] 10.252.249.30 (10.252.249.30)  34.306 ms
    [AS0] 10.252.235.94 (10.252.235.94)  28.505 ms
 8  [AS15169] 74.125.245.209 (74.125.245.209)  25.037 ms
    [AS15169] 172.253.65.248 (172.253.65.248)  25.233 ms
    [AS15169] 74.125.245.209 (74.125.245.209)  30.081 ms
 9  [AS15169] 74.125.245.195 (74.125.245.195)  23.816 ms
    [AS15169] 74.125.245.212 (74.125.245.212)  23.487 ms
    [AS15169] 74.125.245.211 (74.125.245.211)  27.122 ms
10  [AS15169] 108.170.229.27 (108.170.229.27)  177.623 ms
    [AS15169] 142.251.49.57 (142.251.49.57)  178.516 ms
    [AS15169] 108.170.229.27 (108.170.229.27)  181.903 ms
11  * [AS15169] 216.239.57.207 (216.239.57.207)  178.690 ms
    [AS15169] 216.239.58.131 (216.239.58.131)  179.055 ms
12  * * [AS15169] 209.85.142.166 (209.85.142.166)  183.934 ms
13  [AS15169] 74.125.37.170 (74.125.37.170)  187.729 ms
    [AS15169] 172.253.66.15 (172.253.66.15)  186.821 ms
    [AS15169] 72.14.236.18 (72.14.236.18)  188.927 ms
14  [AS15169] 216.239.56.183 (216.239.56.183)  222.139 ms
    [AS15169] 66.249.95.123 (66.249.95.123)  309.484 ms *
15  [AS15169] 108.170.232.251 (108.170.232.251)  316.469 ms
    [AS15169] 74.125.253.109 (74.125.253.109)  328.635 ms
    [AS15169] 74.125.253.94 (74.125.253.94)  217.749 ms
16  [AS15169] 172.253.51.189 (172.253.51.189)  217.978 ms
    [AS15169] 172.253.51.187 (172.253.51.187)  216.735 ms
    [AS15169] 172.253.51.219 (172.253.51.219)  217.763 ms
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  [AS0] lk-in-f147.1e100.net (173.194.220.147)  222.778 ms *  215.848 ms
```

UCT network reaches Google.com in **11 hops** while
Afrihost CPT reaches Google.com in **26 hops**

Reading



Computer Networking : Principles, Protocols and Practice
https://4ed.computer-networking.info/syllabus/default/index.html



Computer Networking: A Top Down Approach
8th edition
Jim Kurose, Keith Ross
Person

Course Overview

- **Chapter 1 The Internet (Week 1)**
- 1.1 What Is the Internet?
- 1.2 The Network Edge
- 1.3 The Network Core
- 1.4 Delay, Loss, and Throughput in Packet-Switched Networks
- 1.5 Protocol Layers and Their Service Models

- **Chapter 2 Application Layer (Week1-2)**
- 2.1 Principles of Network Applications
- 2.7 Socket Programming: Creating Network Applications
- 2.2 The Web and HTTP
- 2.4 DNS—The Internet's Directory Service
- 2.6 Video Streaming and Content Distribution Networks

- **Chapter 3 Transport Layer (Week 3)**
- 3.1 Introduction and Transport-Layer Services
- 3.2 Multiplexing and Demultiplexing
- 3.4 Principles of Reliable Data Transfer
- 3.5 Connection-Oriented Transport: TCP
- 3.6 Principles of Congestion Control

- **Chapter 4 The Network Layer: Data Plane (Week 4)**
- 4.1 Overview of Network Layer
- 4.3 The Internet Protocol (IP): IPv4, Addressing, IPv6

- **Chapter 5 The Network Layer: Control Plane (Week 4)**
- 5.1 Introduction
- 5.2 Routing Algorithms 410 (only LS & DV)
- 5.4 Routing Among the ISPs: BGP
- 5.5 The SDN Control Plane

- **Chapter 6/7 The Link Layer and LANs (Week 5)**
- 6.1 Introduction to the Link Layer
- 6.4 Switched Local Area Networks
- 7.3 WiFi: 802.11 Wireless LANs

- **Chapter 8 Security (Week 6)**
- 8.1 What Is Network Security?
- 8.2 Principles of Cryptography
- 8.3 Message Integrity and Digital Signatures
- 8.4/.5 End-Point Authentication
- 8.6 Securing TCP connections: TLS
- 8.8 Security in wireless and mobile networks
- 8.9 Operational security: firewalls and IDS

Assessment

Practical Assignment 1: Socket programming (Group Assignment)

- Open: Thursday, 19 Feb
- Stage1 submission: Thursday, 26 Feb
- Stage2 submission: Thursday, 5 March
- **Stage3 submission: Friday, 13 March**

Practical Assignment 2: Traffic Analysis (Individual, Closed Prac)

* Wireshark, Linux net-tools (e.g Ping, Traceroute, Dig, Whois, nslookup, etc)

- Open Practice: Available 6 March
- **Closed Practical: 17, 18 March**

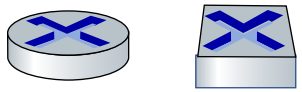
Class Test: *25 Mar 6:00PM Hanh LT1, LT2, LT3, John Day LT1*

The Internet: a "nuts and bolts" view



Billions of connected computing *devices*:

- *hosts = end systems*
- running *network apps* at Internet's "edge"



Packet switches: forward packets (chunks of data)

- *routers, switches*

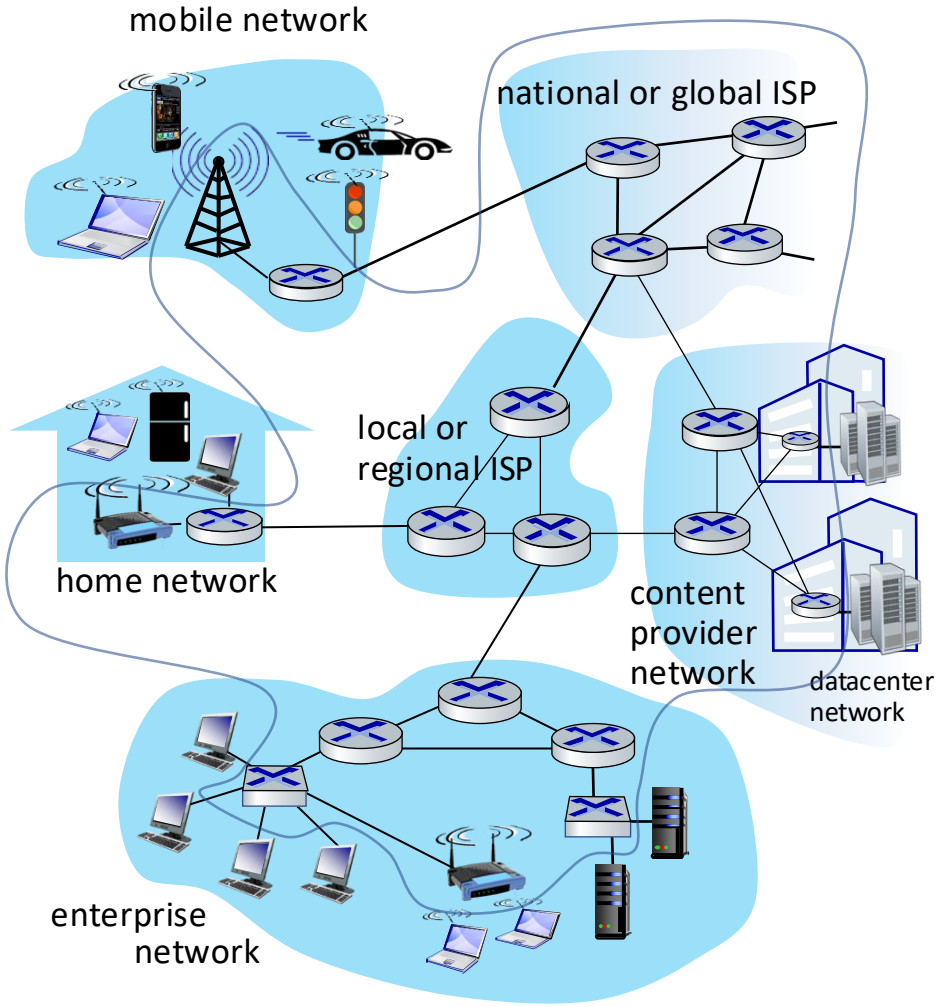


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

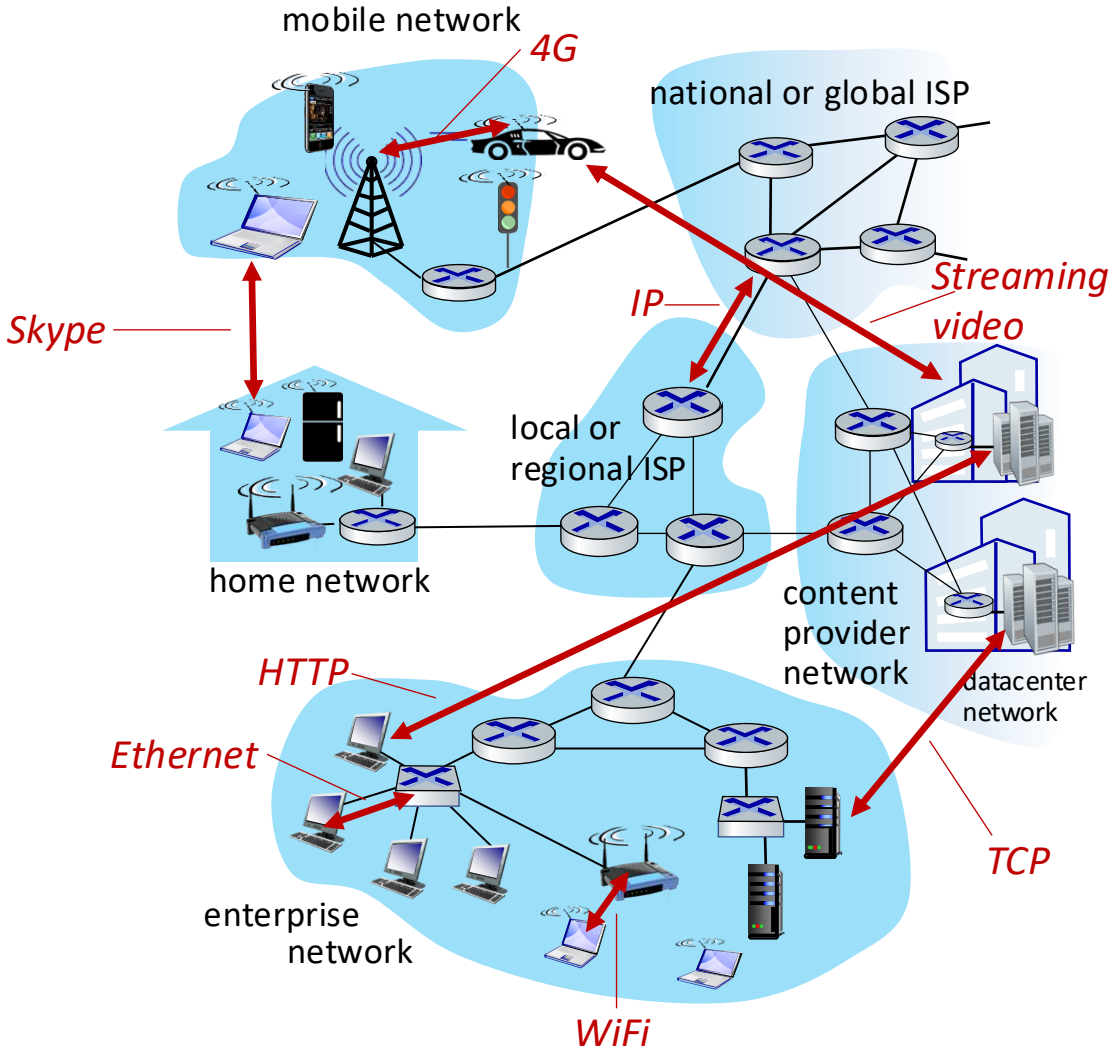
Networks

- collection of devices, routers, links: managed by an organization



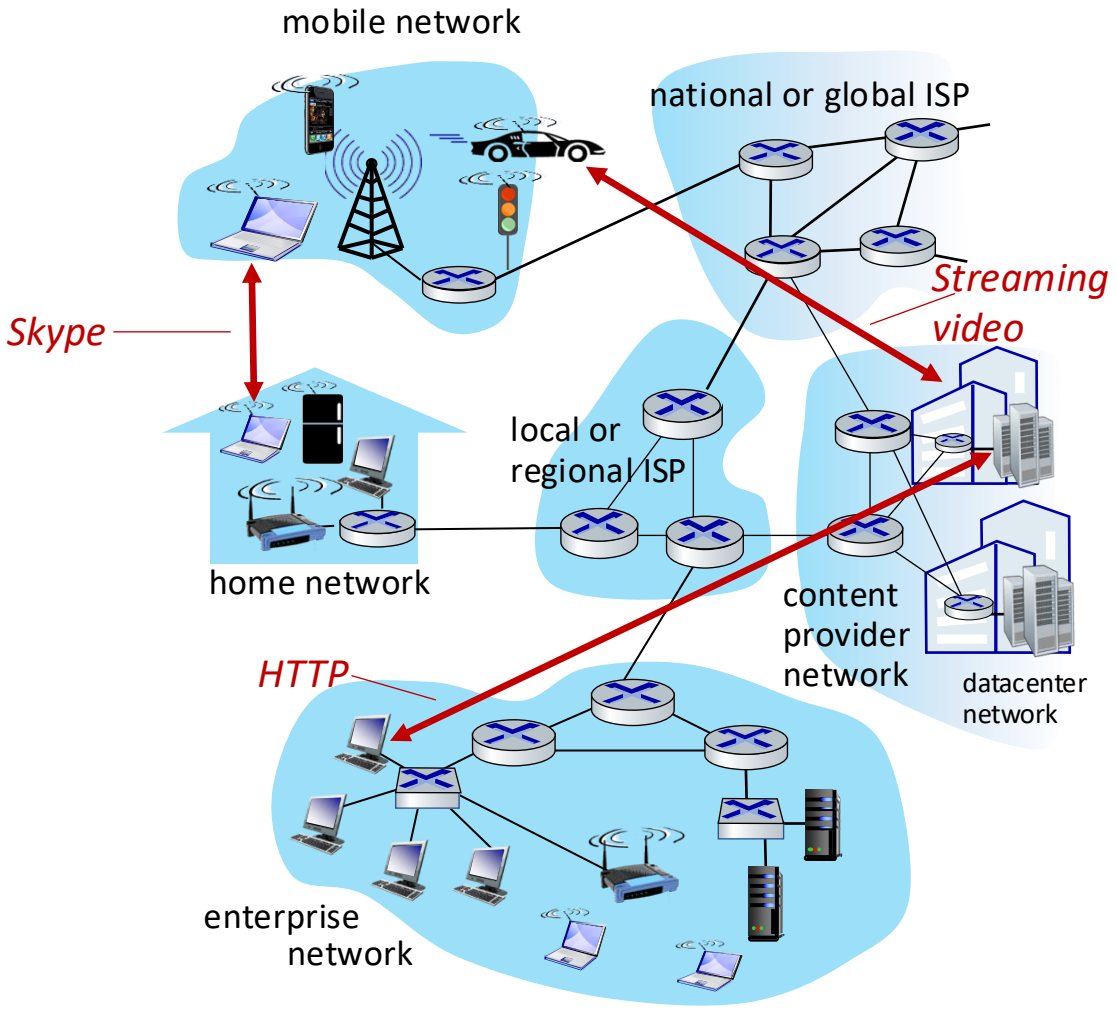
The Internet: a "nuts and bolts" view

- *Internet:* "network of networks"
  - Interconnected ISPs
- *protocols* are *everywhere*
  - control sending, receiving of messages
  - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet
- *Internet standards*
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force



The Internet: a "services" view

- *Infrastructure* that provides services to applications:
  - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, …
- provides *programming interface* to distributed applications:
  - "hooks" allowing sending/receiving apps to "connect" to, use Internet transport service
  - provides service options, analogous to postal service



Roadmap

- What *is* the Internet?
- Protocol layers, service models
- What *is* a protocol?
- Network edge: hosts, access network
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput



Protocol “layers” and reference models

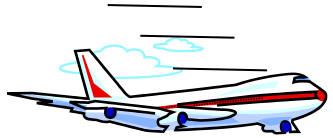
Networks are complex, with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any hope of *organizing* structure of network?

- and/or our *discussion* of networks?

Example: organization of air travel



end-to-end transfer of person plus baggage →

ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

runway landing

airplane routing

airplane routing

How would you *define/discuss* the *system* of airline travel?

- a series of steps, involving many services

Example: organization of air travel



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

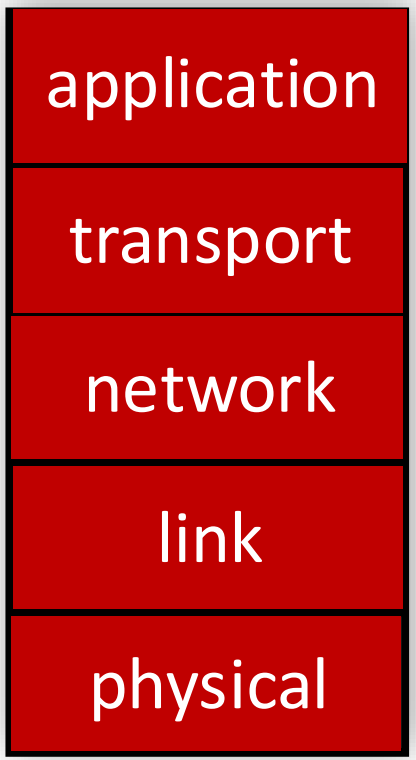
Why layering?

Approach to designing/discussing complex systems:

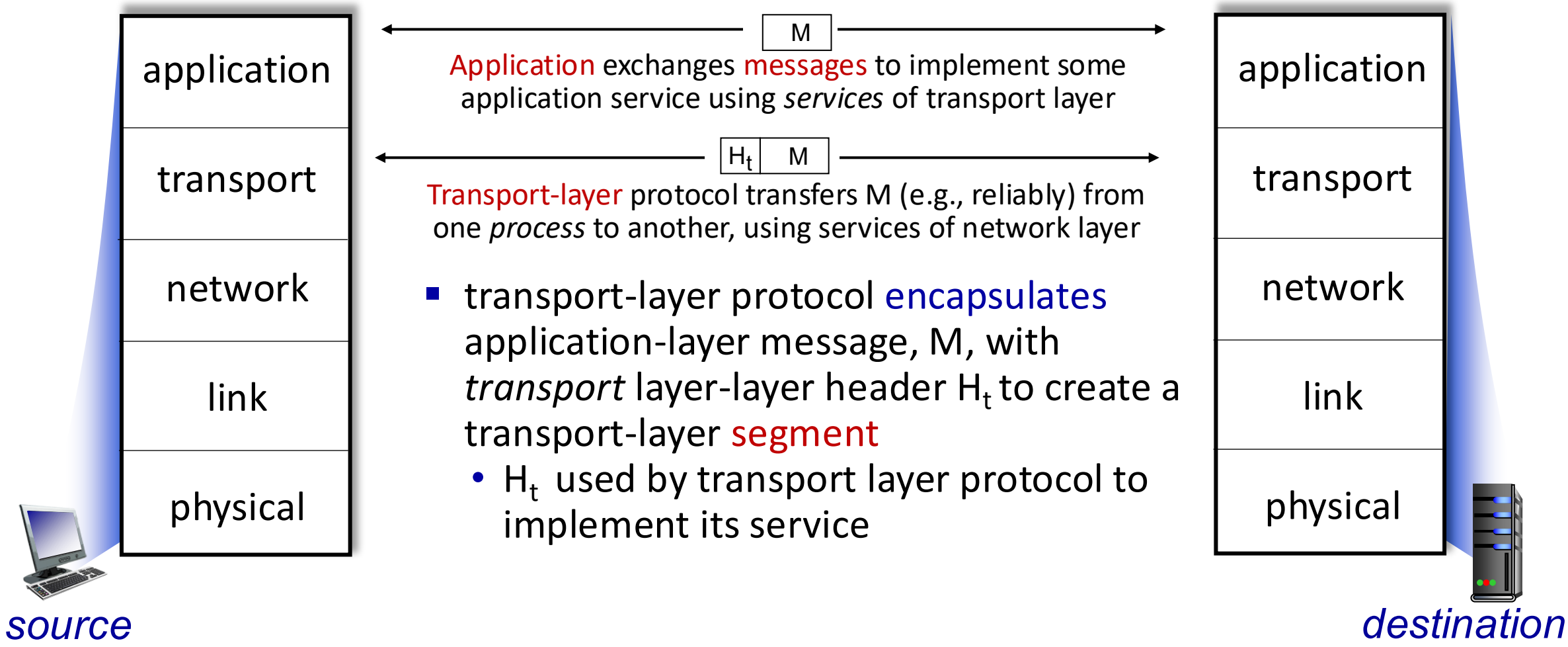
- explicit structure allows identification, relationship of system's pieces
  - layered *reference model* for discussion
- modularization eases maintenance, updating of system
  - change in layer's service *implementation*: transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system

Layered Internet protocol stack

- *application:* supporting network applications
  - HTTP, IMAP, SMTP, DNS
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.11 (WiFi), PPP
- *physical:* bits “on the wire”

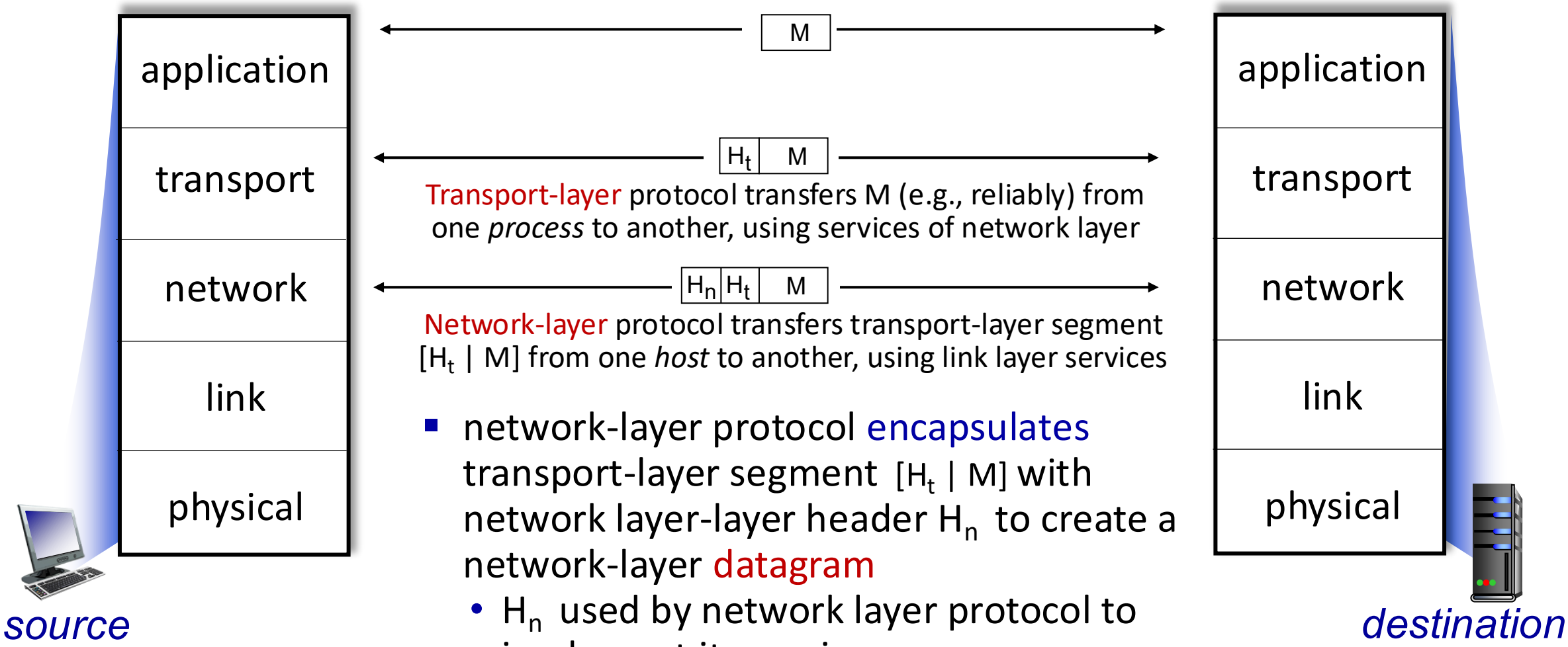


Services, Layering and Encapsulation



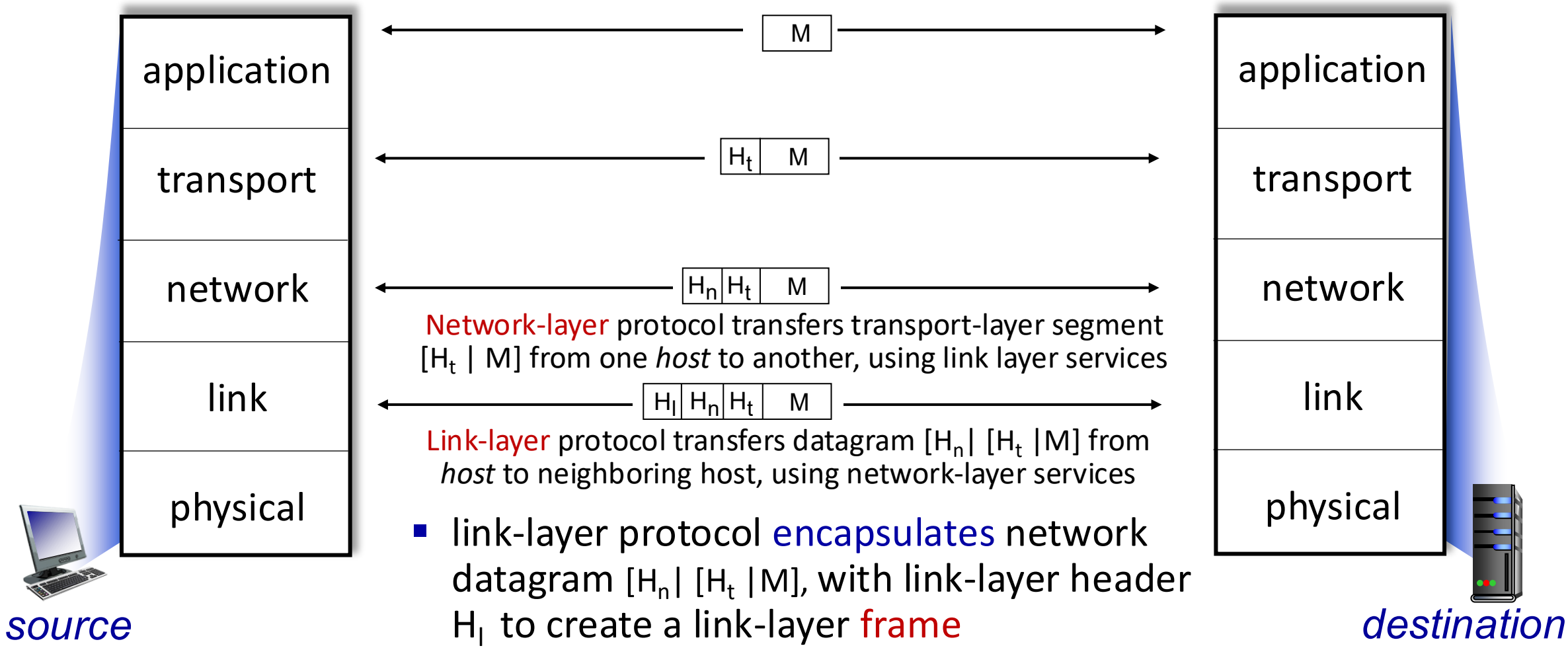
- transport-layer protocol encapsulates application-layer message, M, with *transport* layer-layer header H_t to create a transport-layer segment
  - H_t used by transport layer protocol to implement its service

Services, Layering and Encapsulation



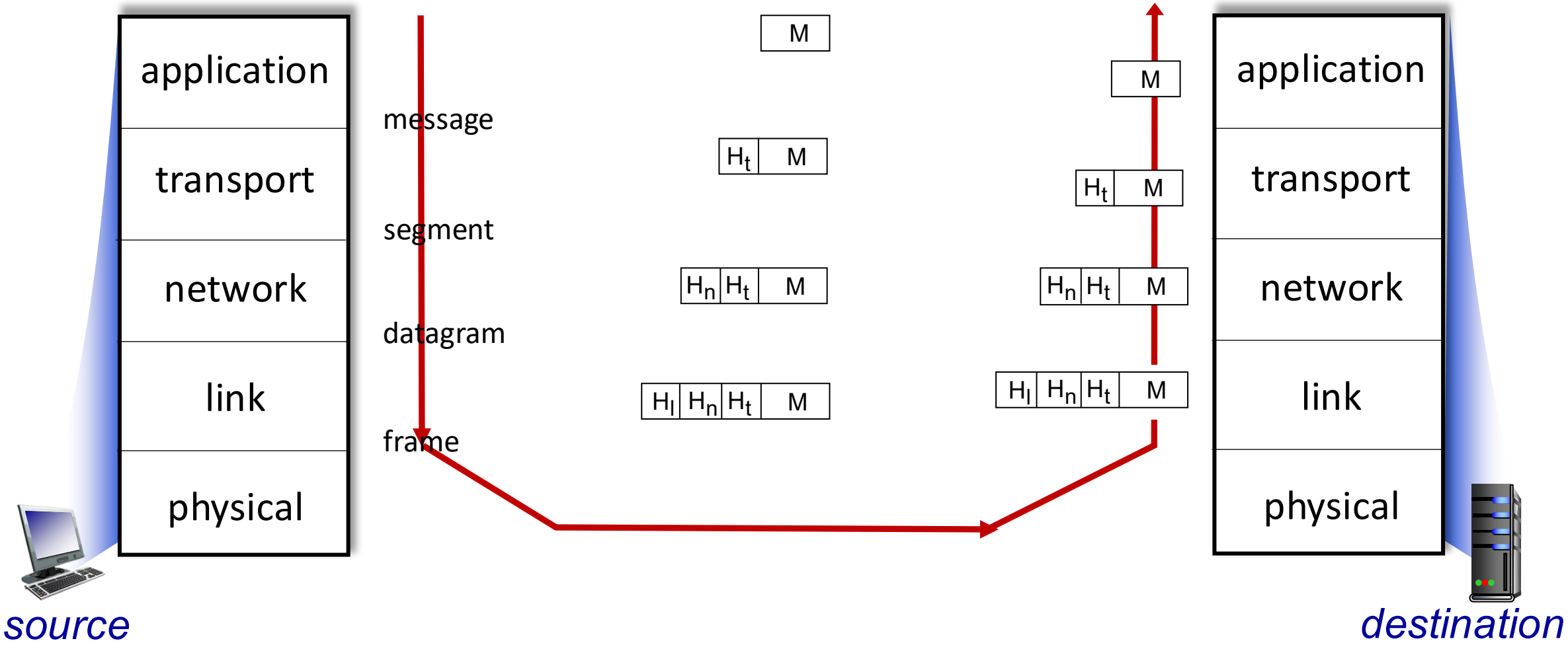
- network-layer protocol encapsulates transport-layer segment [H_t | M] with network layer-layer header H_n to create a network-layer datagram
  - H_n used by network layer protocol to implement its service

Services, Layering and Encapsulation



- link-layer protocol encapsulates network datagram [H_n| [H_t |M], with link-layer header H_l to create a link-layer frame

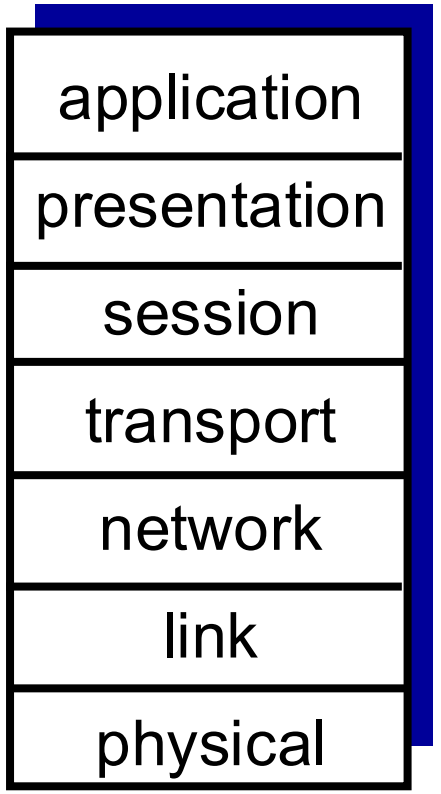
Services, Layering and Encapsulation



ISO/OSI reference model

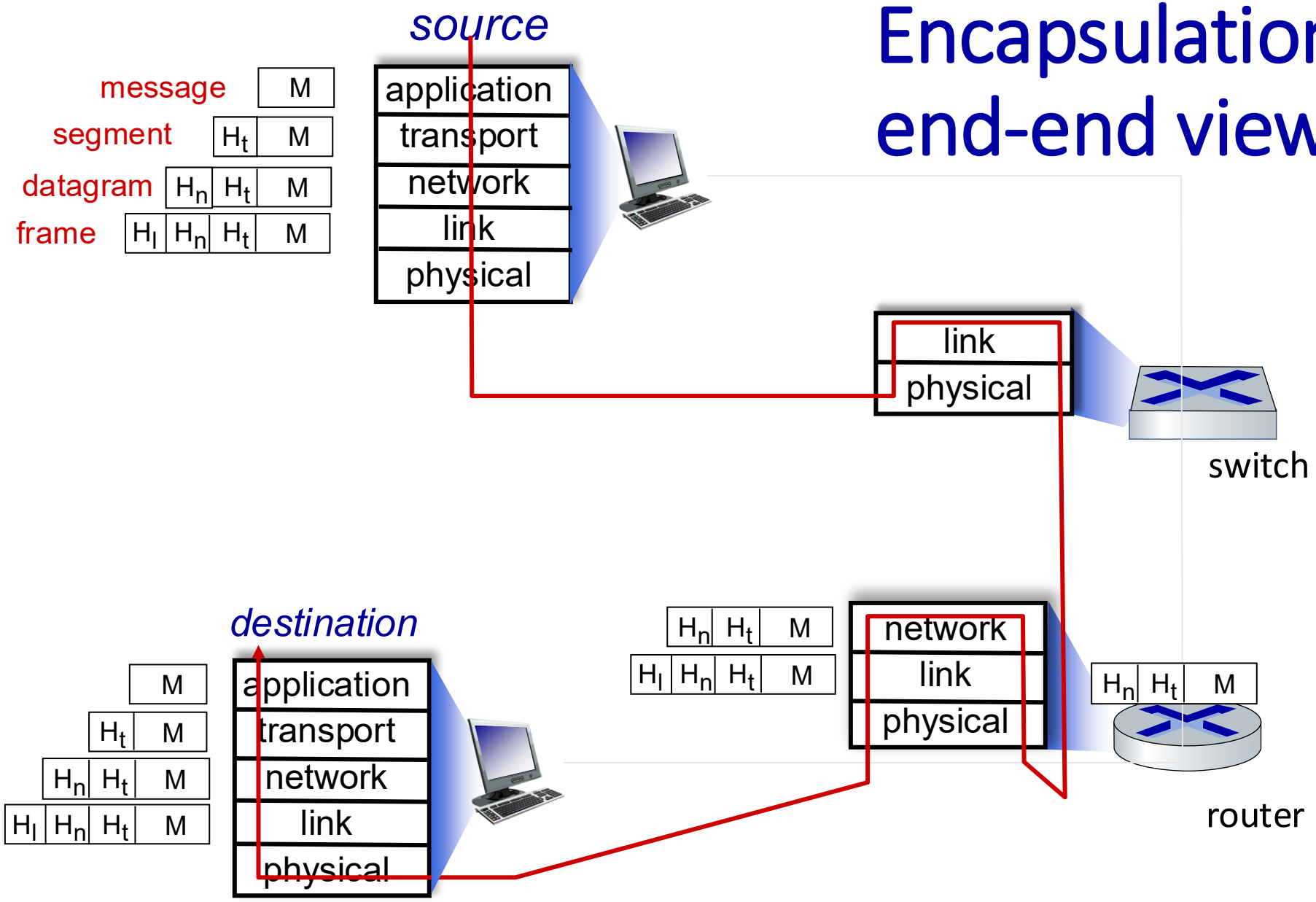
Two layers not found in Internet protocol stack!

- *presentation:* allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- *session:* synchronization, checkpointing, recovery of data exchange
- Internet stack "missing" these layers!
  - these services, *if needed,* must be implemented in application
  - needed?



The seven layer OSI/ISO reference model

Encapsulation: an end-end view



Roadmap

- What *is* the Internet?
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What's a protocol?

Human protocols:

- "what's the time?"
- "I have a question"
- introductions

Rules for:

… specific messages sent

… specific actions taken when message received, or other events

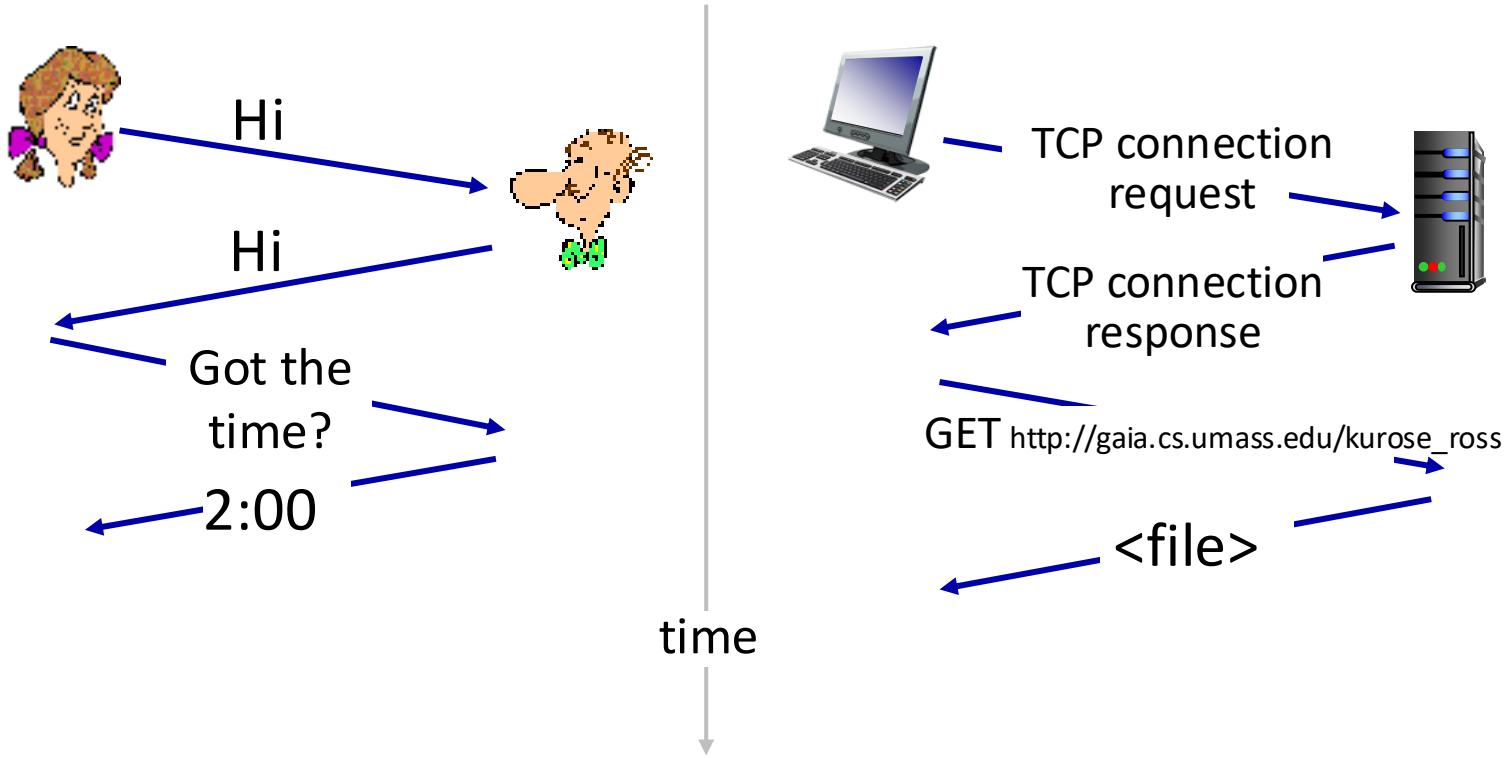
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Protocols define the format, order of messages sent and received among network entities, and actions taken on message transmission, receipt

What's a protocol?

A human protocol and a computer network protocol:



Q: other human protocols?

Roadmap

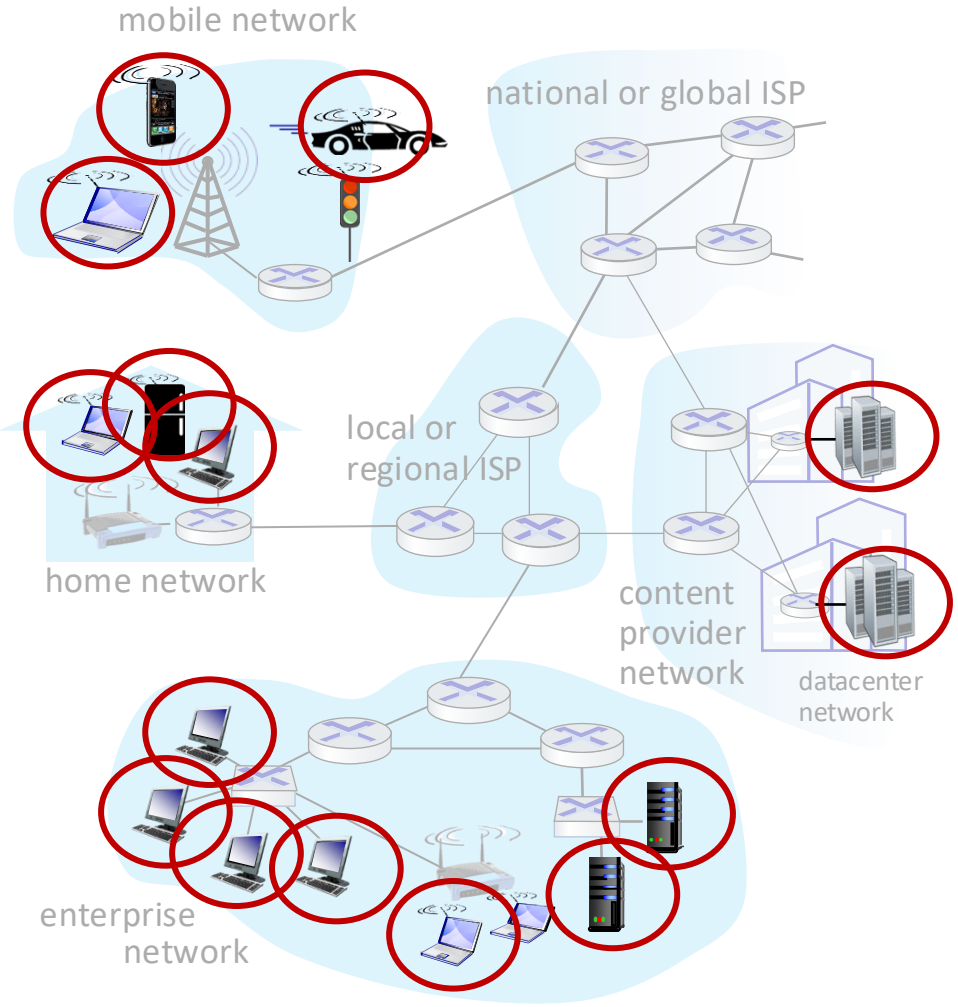
- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network
- Network core: packet/circuit switching, internet structure
- Protocol layers, service models



A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



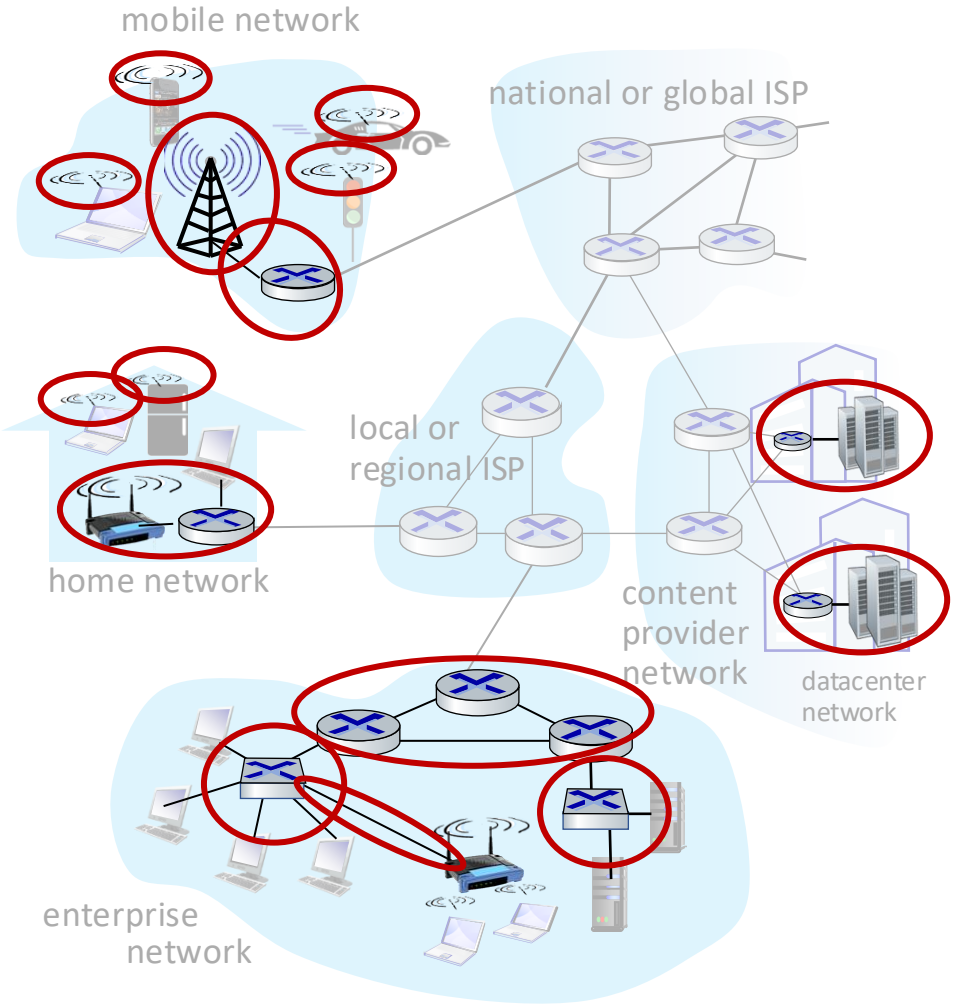
A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links



A closer look at Internet structure

Network edge:

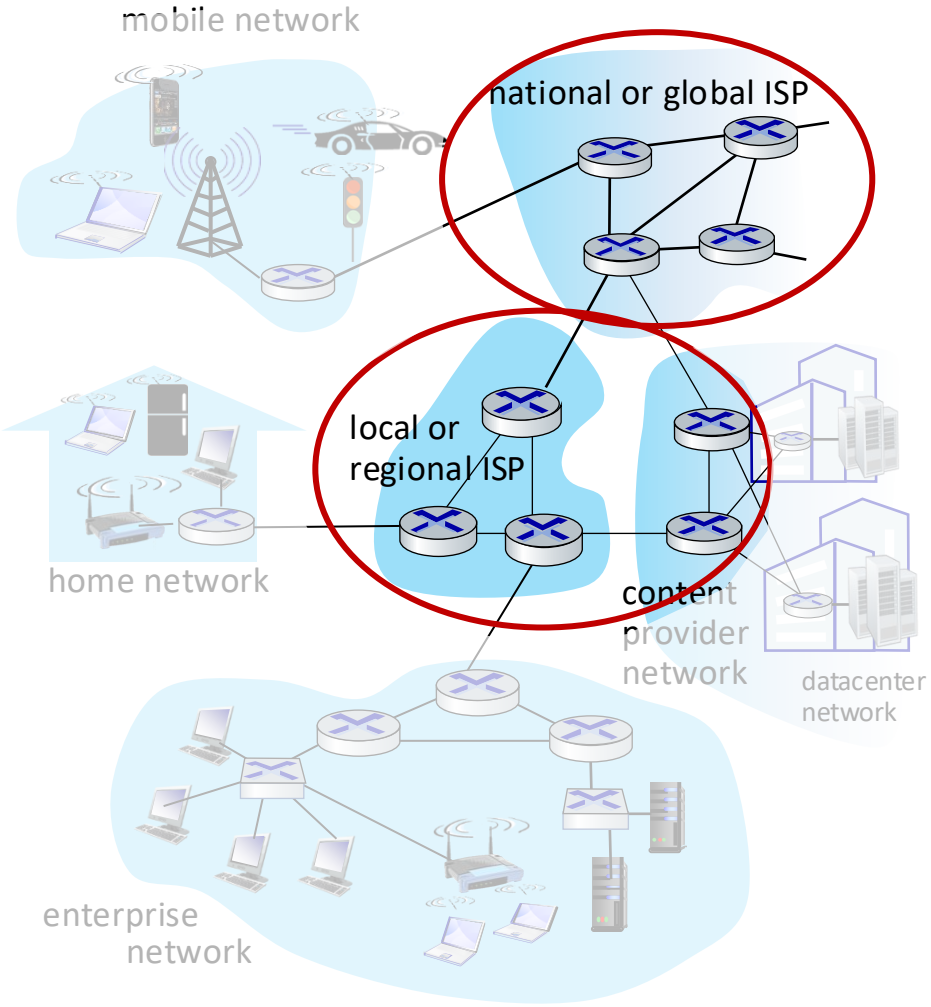
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Access networks, physical media:

- wired, wireless communication links

Network core:

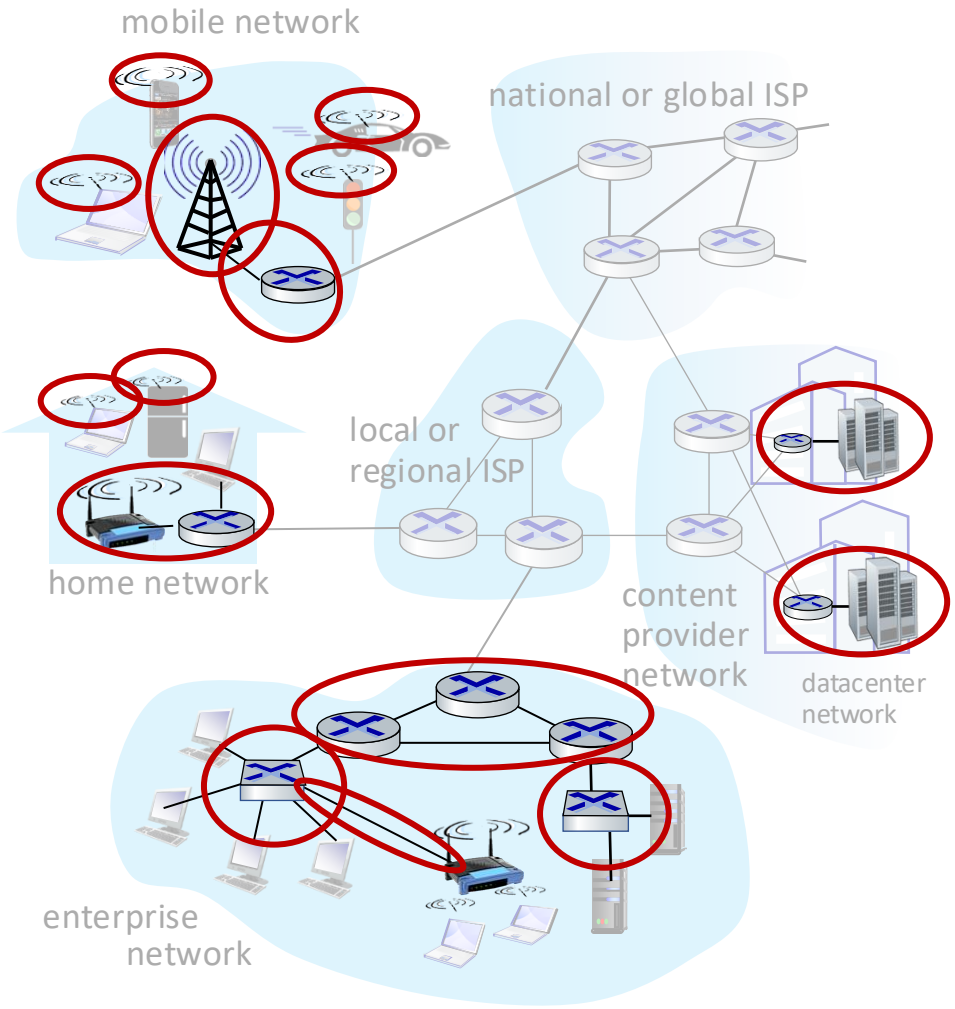
- interconnected routers
- network of networks



Access networks and physical media

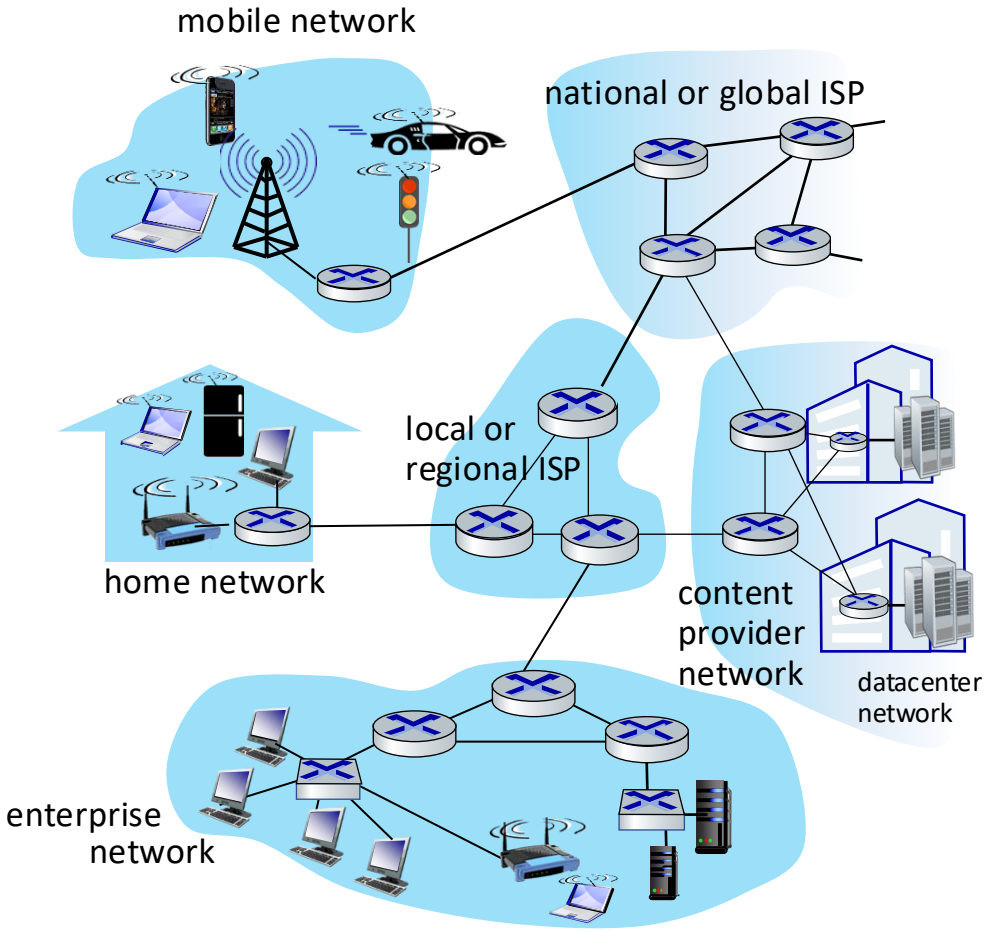
Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)



Internet structure: a "network of networks"

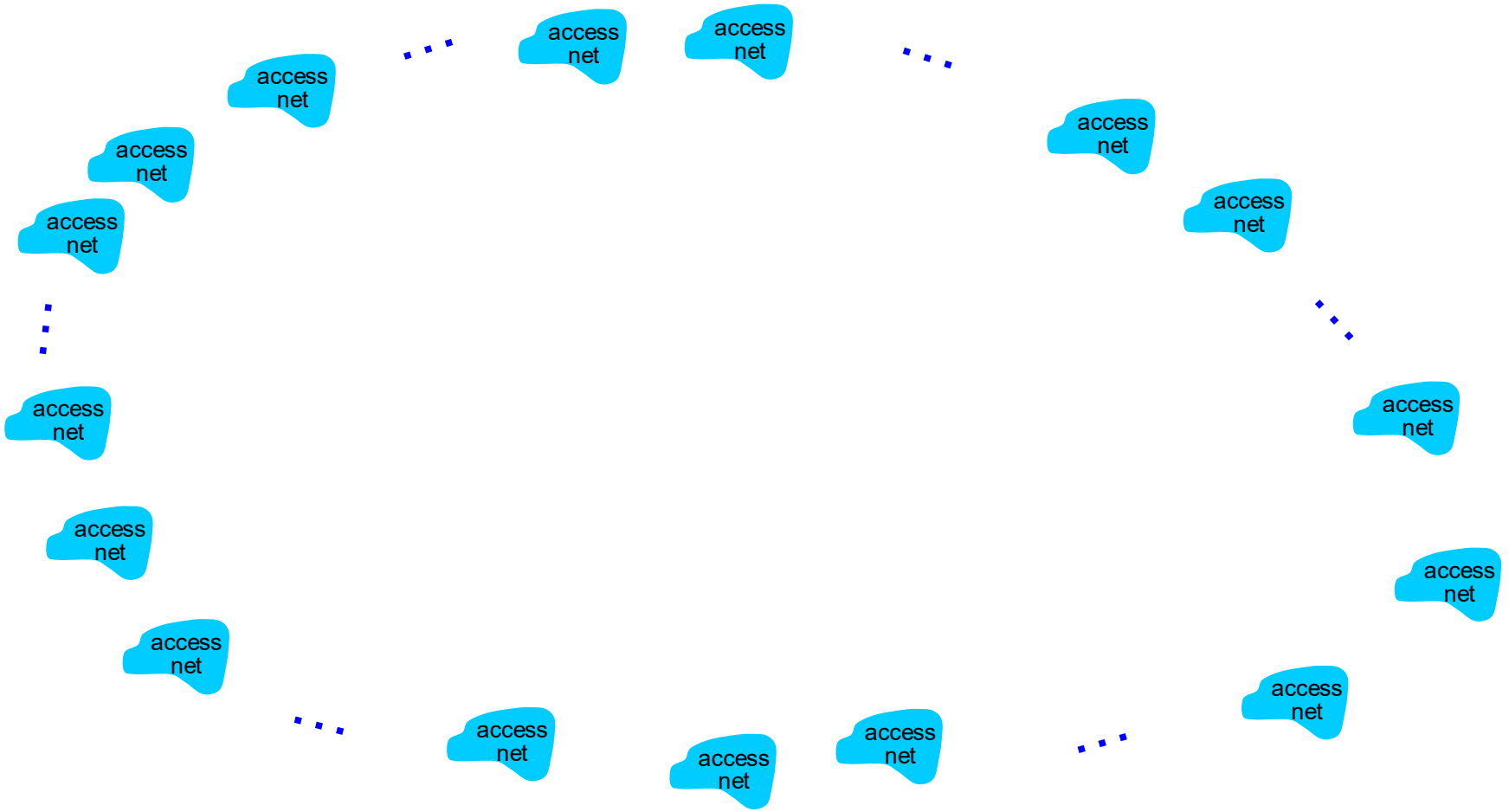
- hosts connect to Internet via access Internet Service Providers (ISPs)
- access ISPs in turn must be interconnected
  - so that *any* two hosts *(anywhere!)* can send packets to each other
- resulting network of networks is very complex
  - evolution driven by economics, national policies



Let's take a stepwise approach to describe current Internet structure

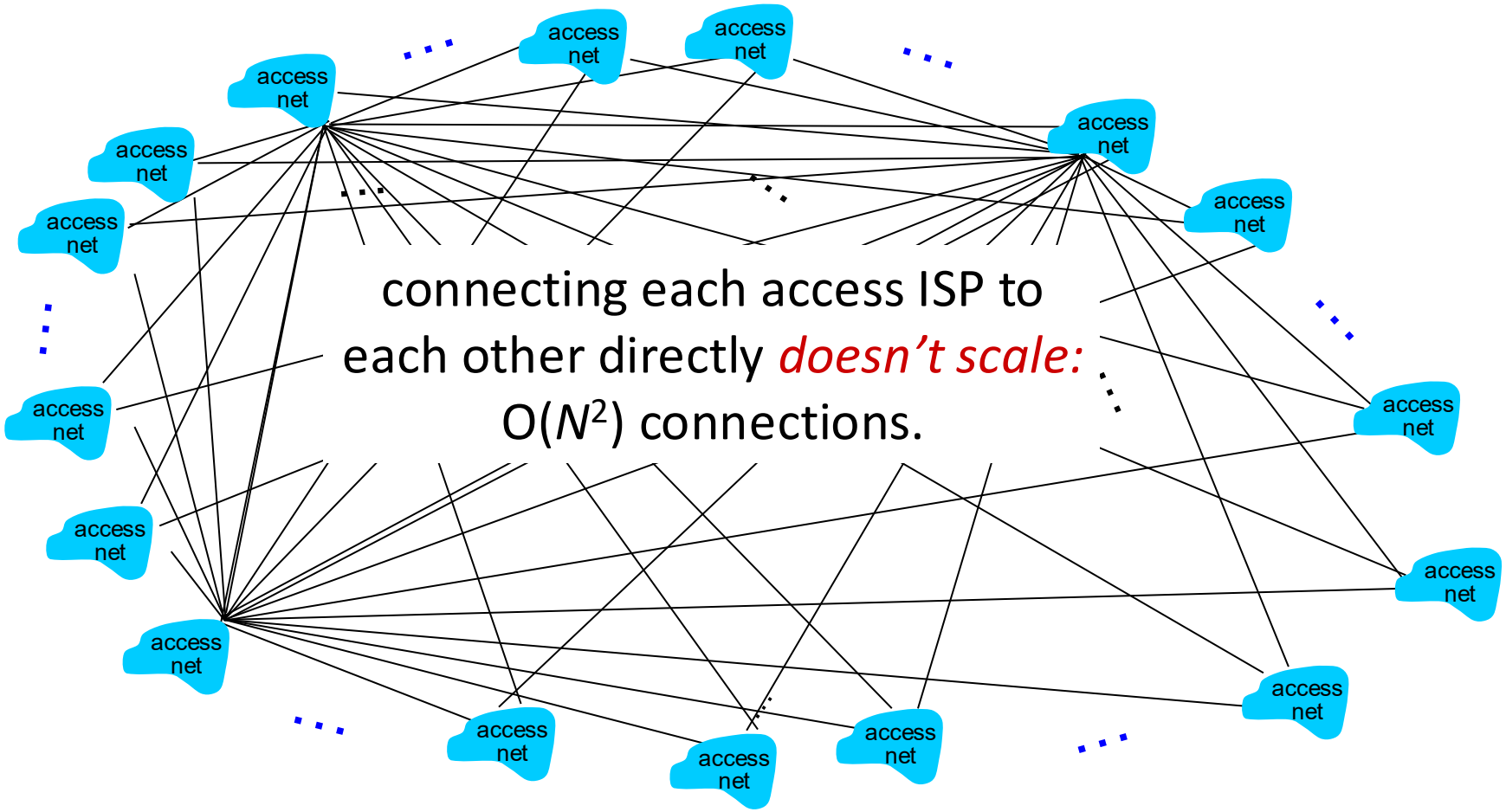
Internet structure: a "network of networks"

Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a "network of networks"

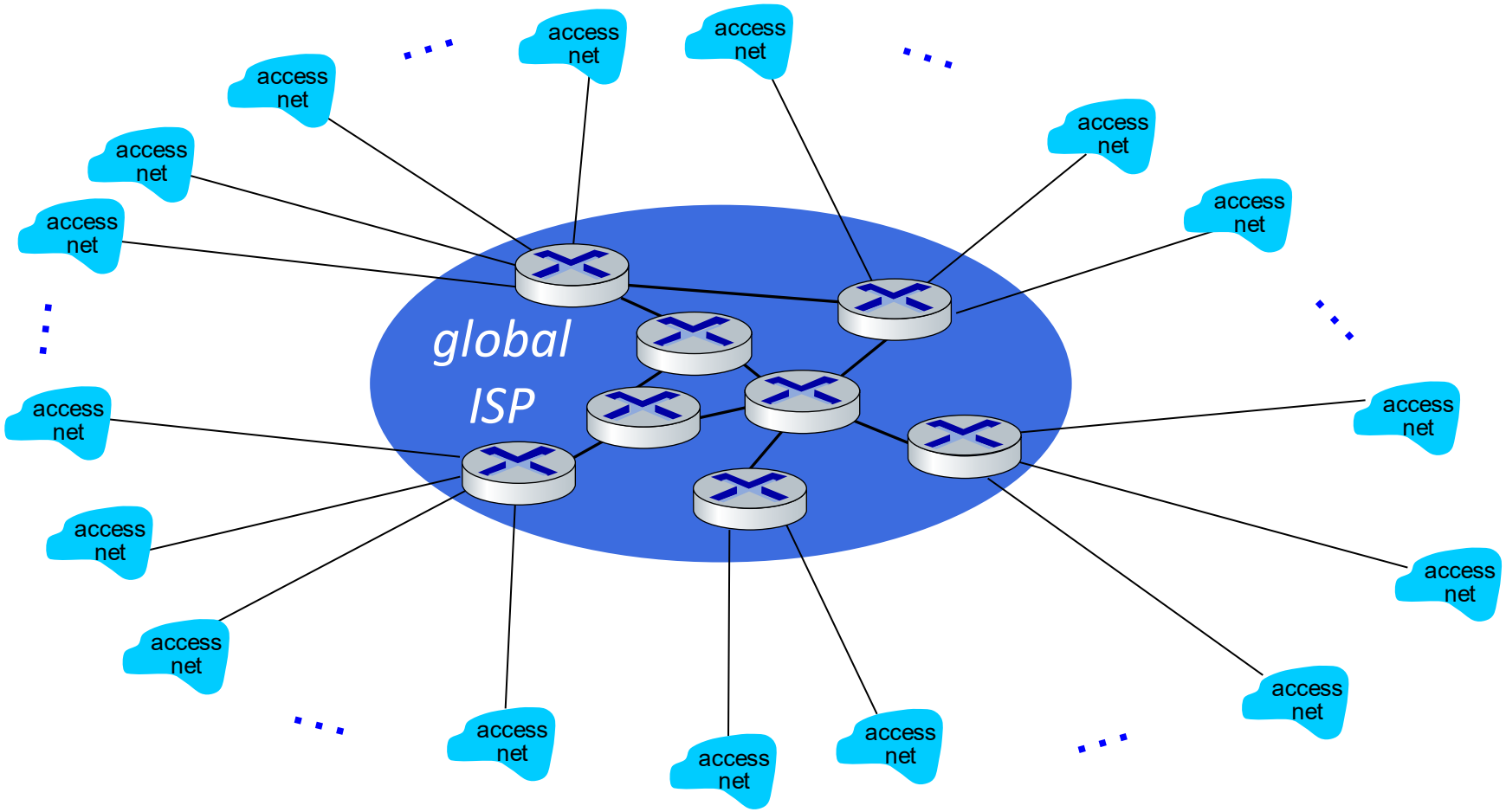
Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a "network of networks"

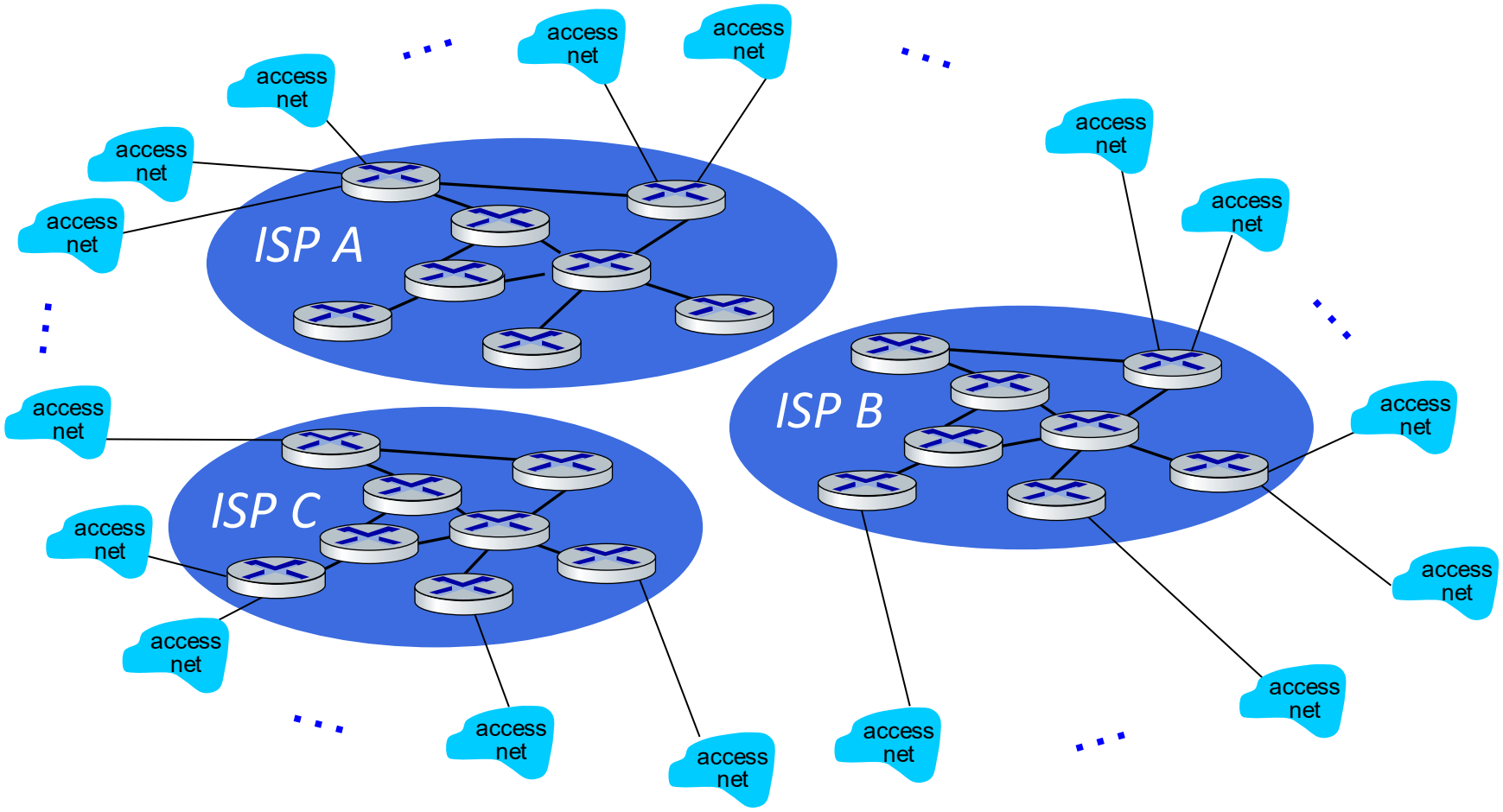
Option: connect each access ISP to one global transit ISP?

Customer and provider ISPs have economic agreement.



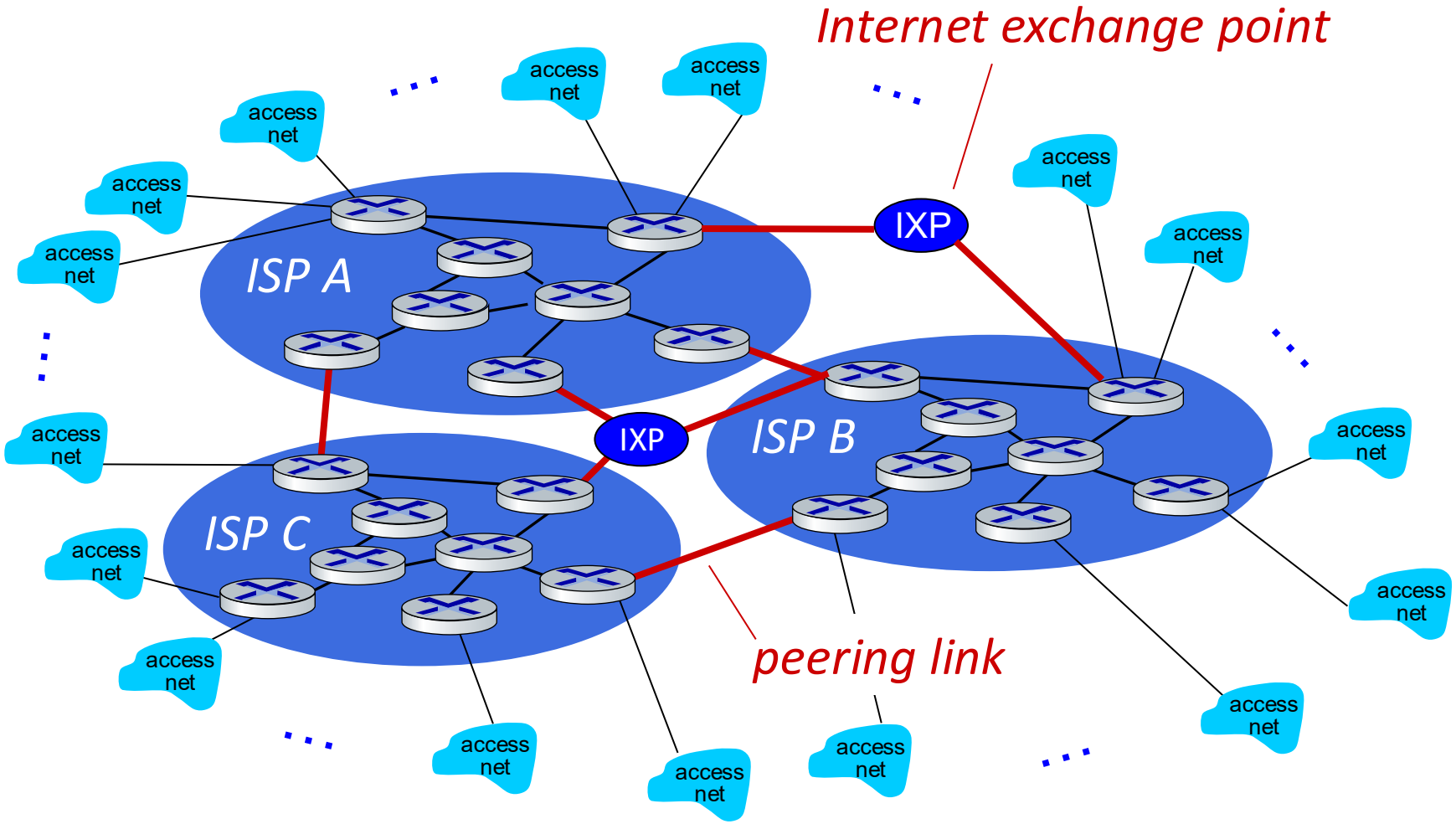
Internet structure: a "network of networks"

But if one global ISP is viable business, there will be competitors ….



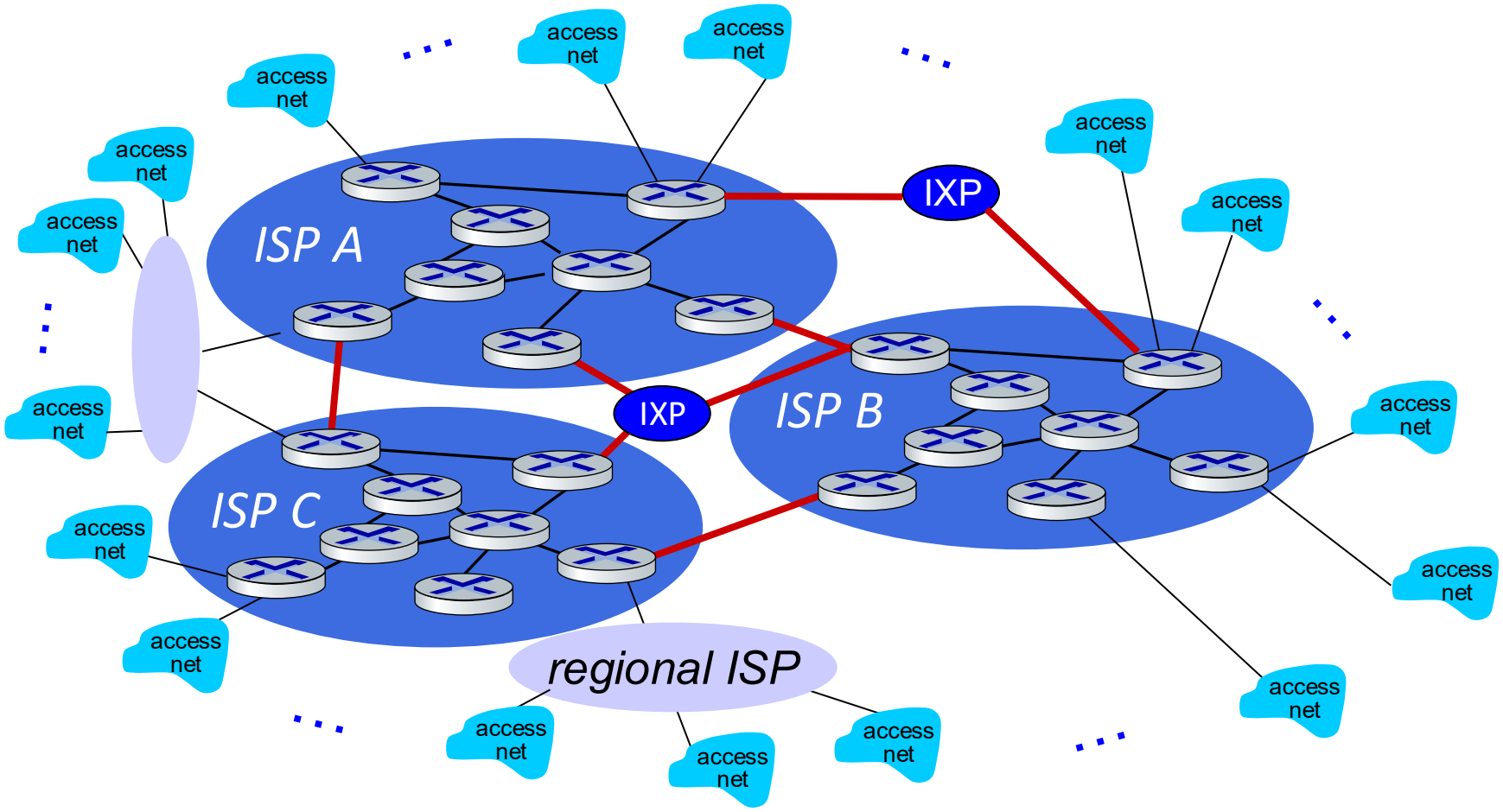
Internet structure: a "network of networks"

But if one global ISP is viable business, there will be competitors …. who will want to be connected



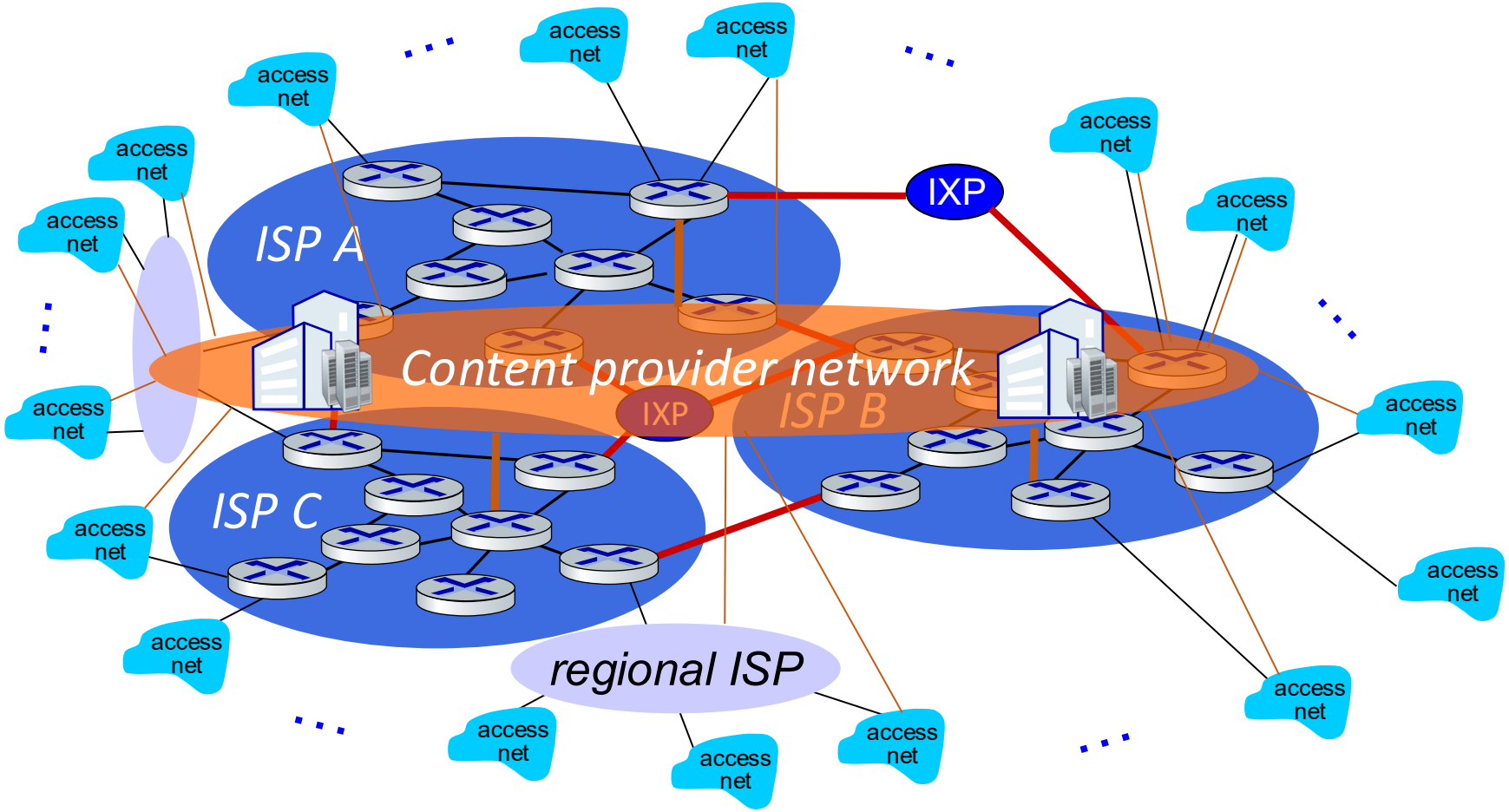
Internet structure: a "network of networks"

… and regional networks may arise to connect access nets to ISPs

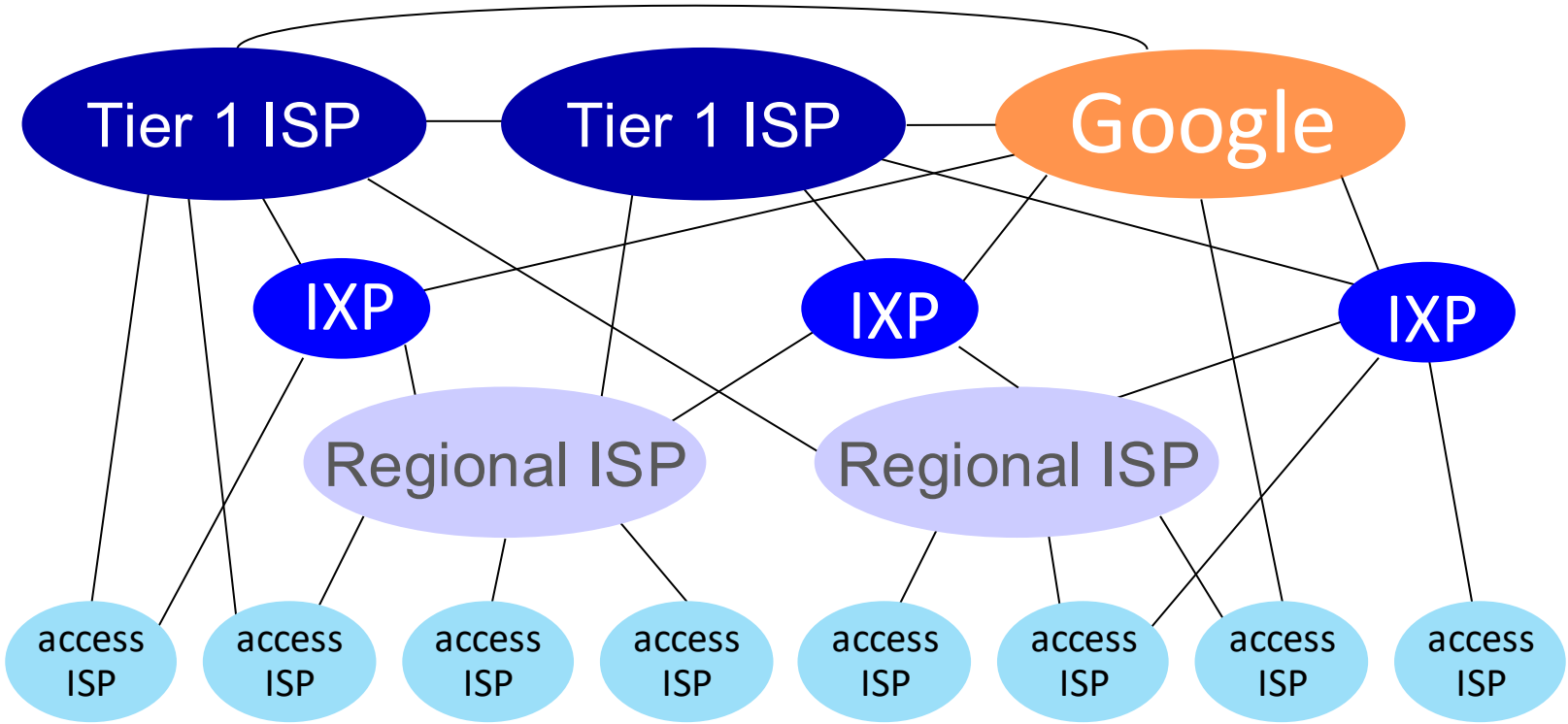


Internet structure: a "network of networks"

… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet structure: a "network of networks"



At "center": small # of well-connected large networks

- "tier-1" commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider networks (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

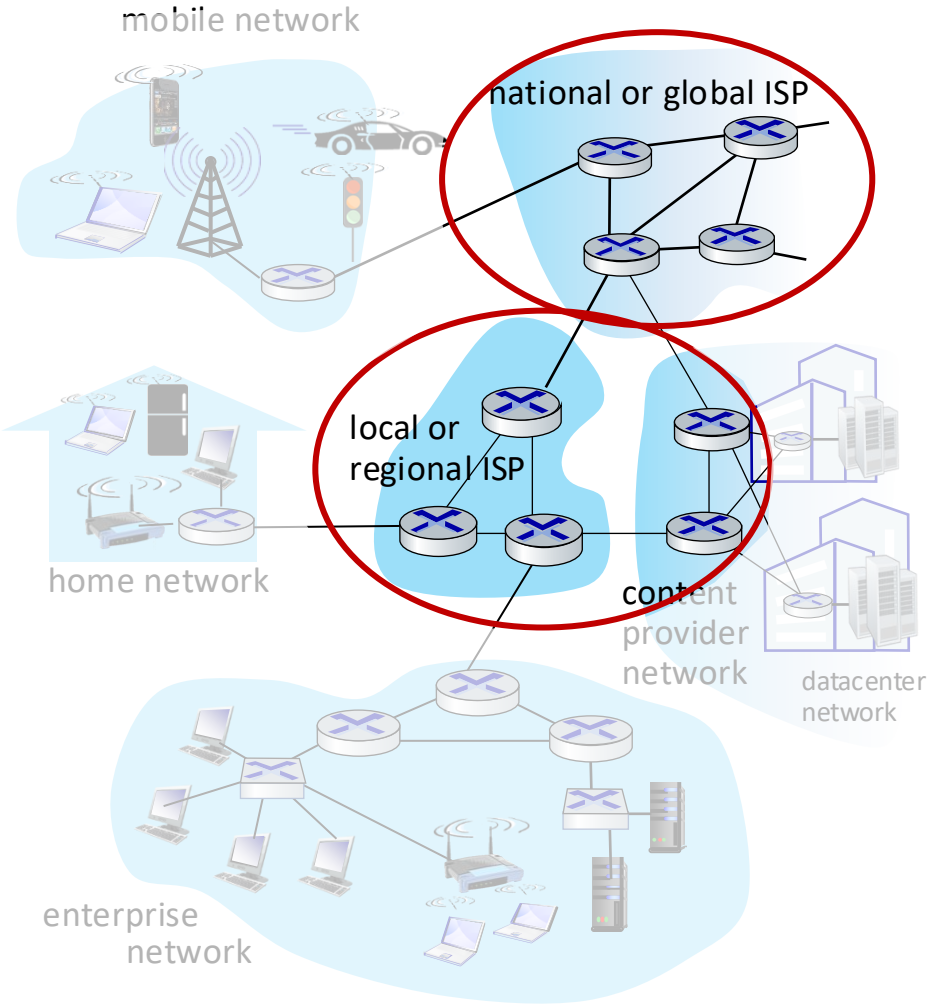
Roadmap

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- What *is* a protocol?
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The network core

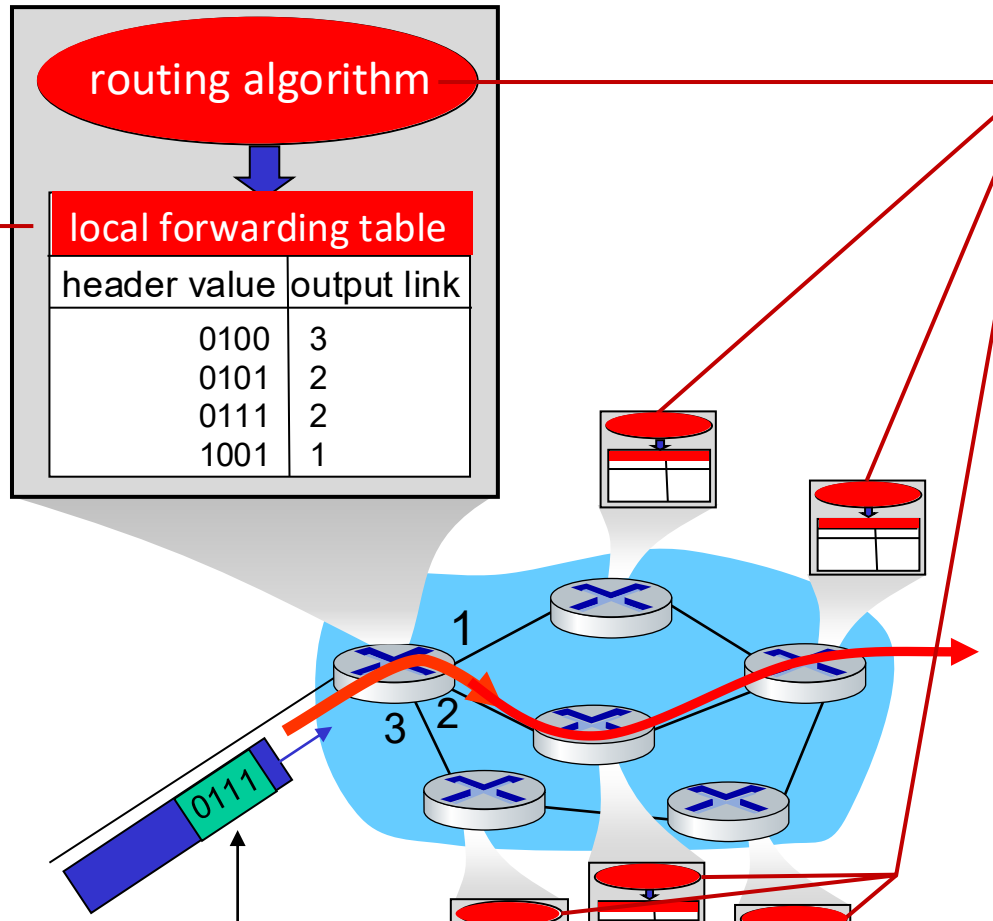
- mesh of interconnected routers
- packet-switching: hosts break application-layer messages into *packets*
  - network forwards packets from one router to the next, across links on path from source to destination



Two key network-core functions

Forwarding:

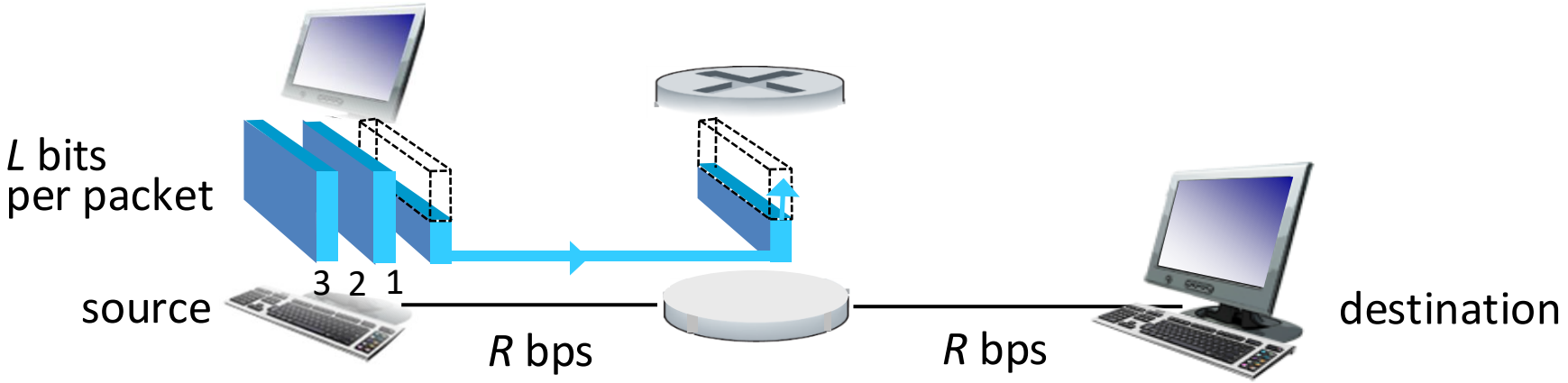
- aka "switching"
- *local* action: move arriving packets from router's input link to appropriate router output link



Routing:

- *global* action: determine source-destination paths taken by packets
- routing algorithms

Packet-switching: store-and-forward

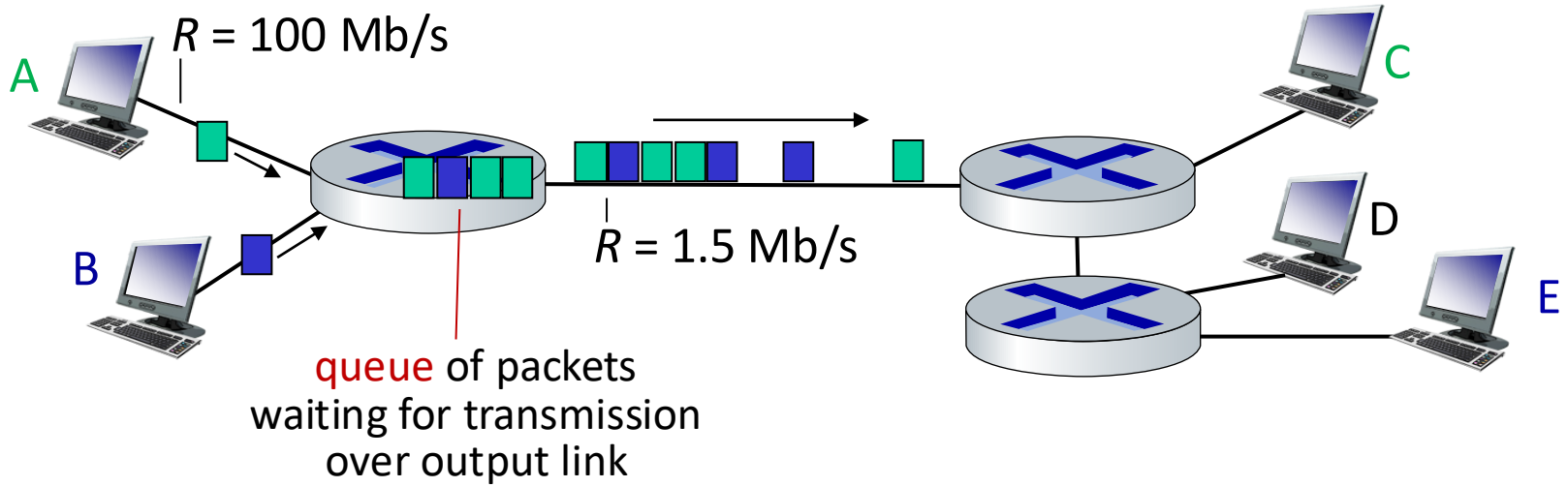


- packet transmission delay: takes L/R seconds to transmit (push out) L-bit packet into link at R bps
- *store and forward: entire* packet must arrive at router before it can be transmitted on next link

One-hop numerical example:

- L = 10 Kbits
- R = 100 Mbps
- one-hop transmission delay = 0.1 msec

Packet-switching: queueing



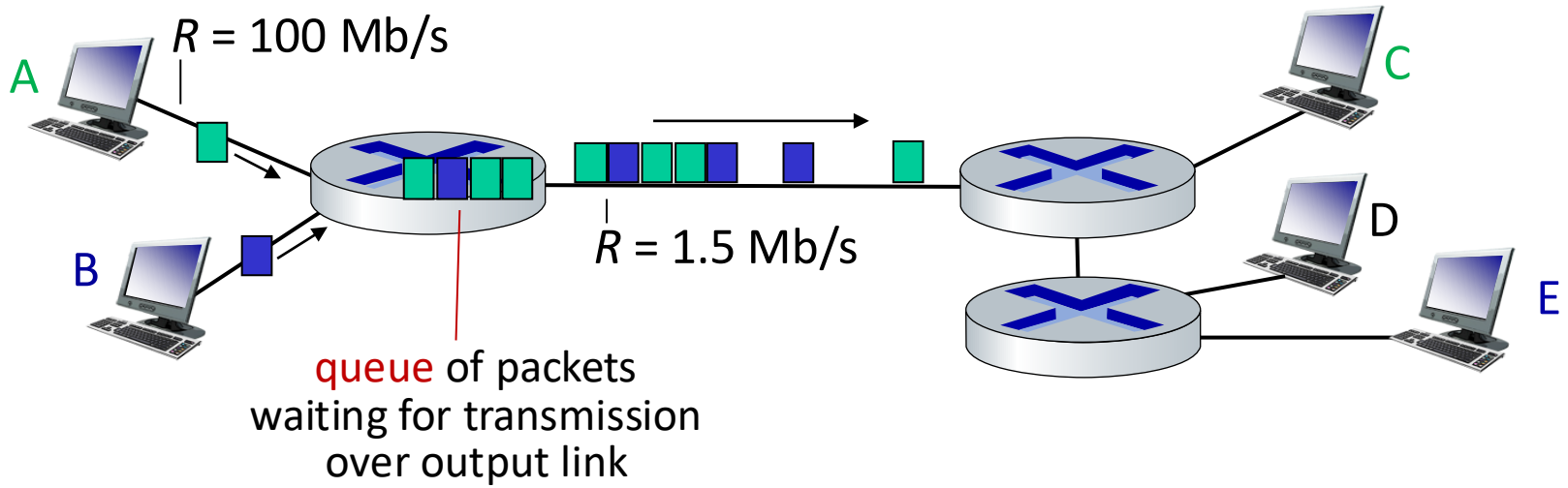
Queueing occurs when work arrives faster than it can be serviced:







Packet-switching: queueing



Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

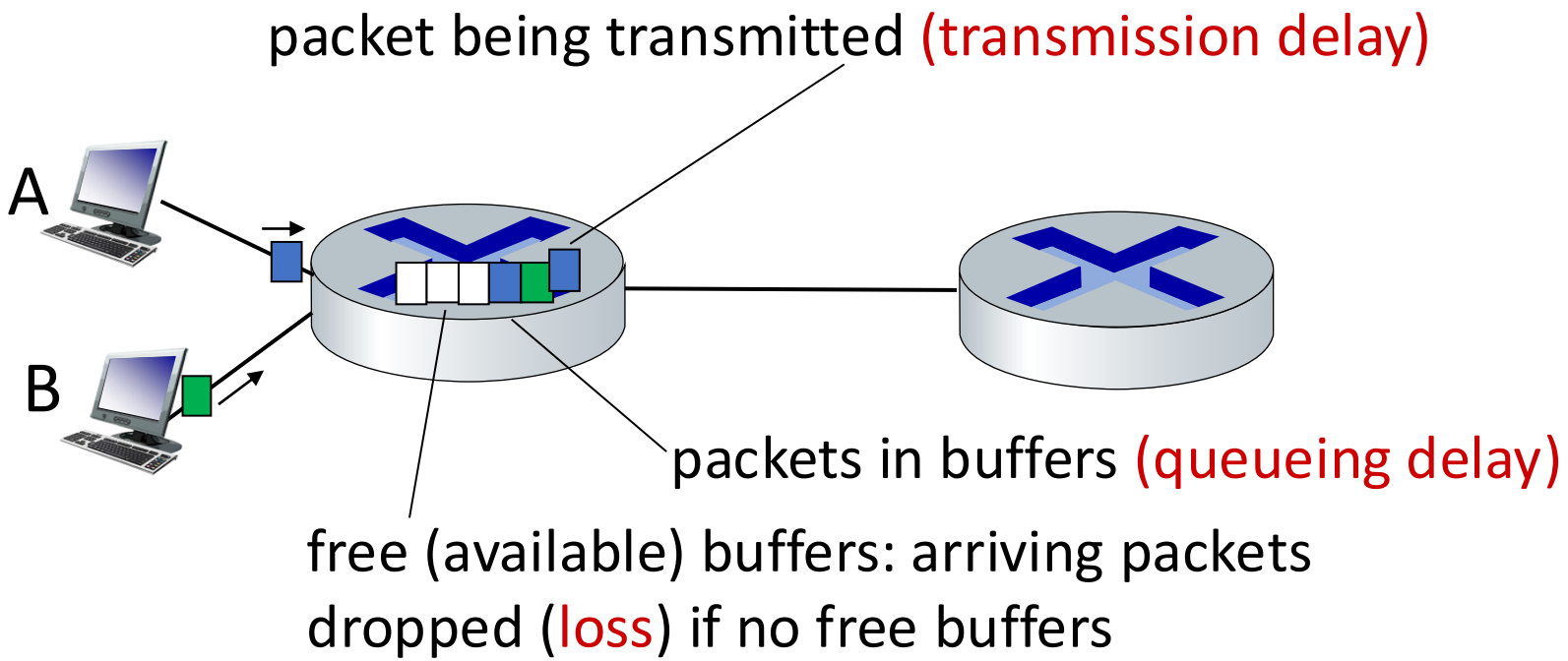
Roadmap

- What *is* the Internet?
- What *is* a protocol?
- Protocol layers, service models
- Network edge: hosts, access network
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput

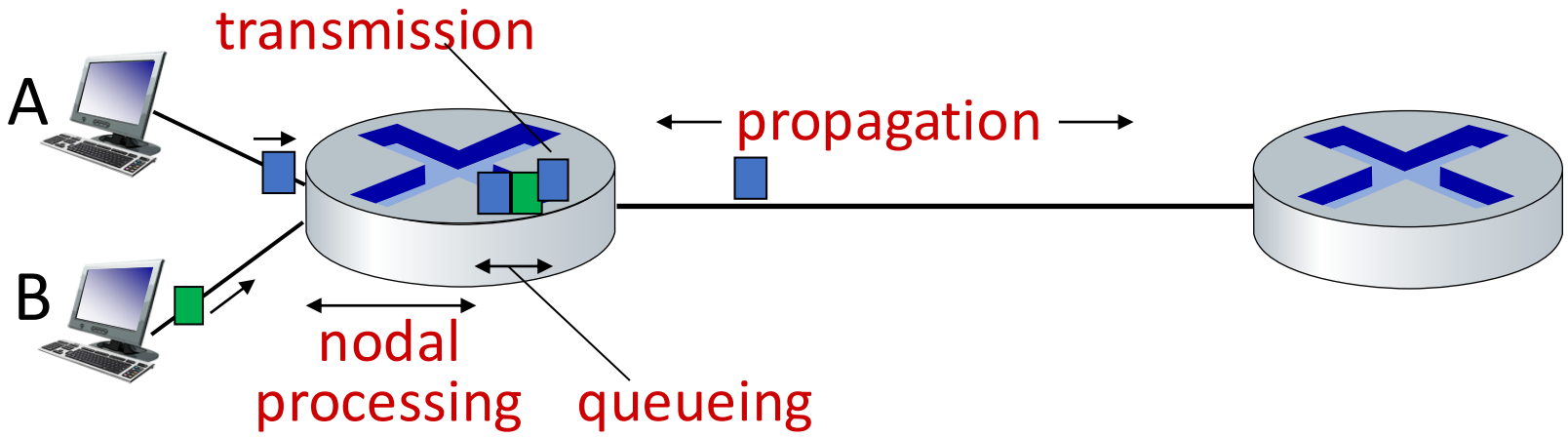


How do packet delay and loss occur?

- packets *queue* in router buffers, waiting for turn for transmission
  - queue length grows when arrival rate to link (temporarily) exceeds output link capacity
- packet *loss* occurs when memory to hold queued packets fills up



Packet delay: four sources



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

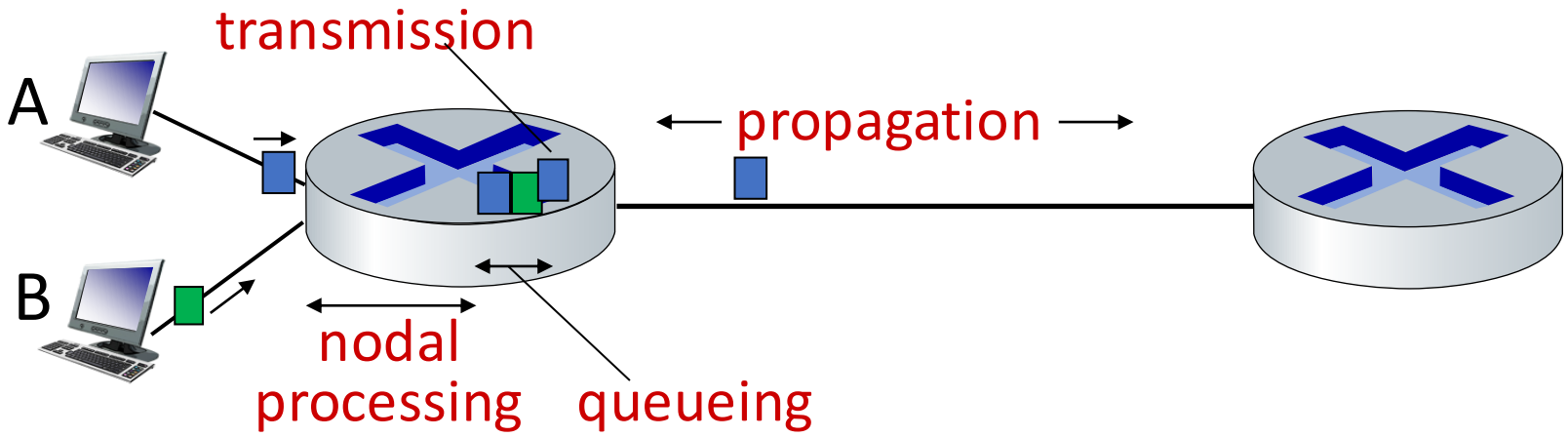
d_{proc}: nodal processing

- check bit errors
- determine output link
- typically < microsecs

d_{queue}: queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Packet delay: four sources



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{trans}: transmission delay:

- L: packet length (bits)
- R: link *transmission rate (bps)*
- $d_{trans} = L/R$

d_{prop}: propagation delay:

- d: length of physical link
- s: propagation speed (~2×10^8 m/sec)
- $d_{prop} = d/s$

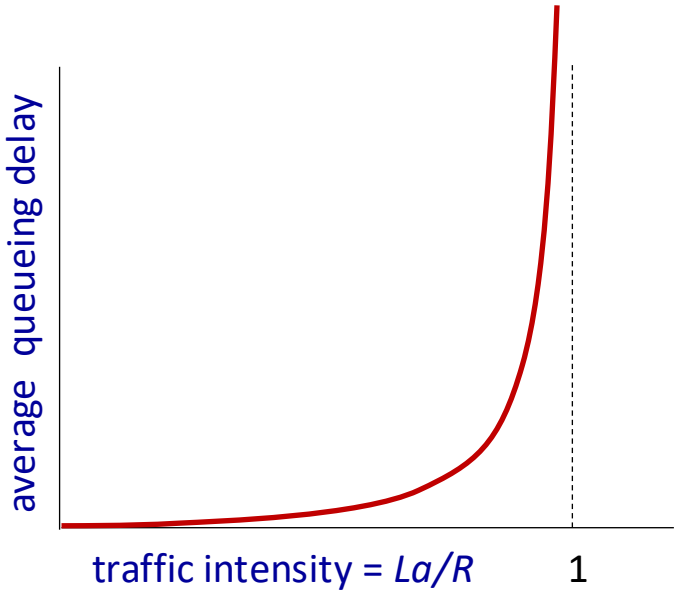
d_{trans} and d_{prop} *very* different

Packet queueing delay (revisited)

- *a:* average packet arrival rate
- *L:* packet length (bits)
- *R:* link bandwidth (bit transmission rate)

$$\frac{L \cdot a}{R} : \frac{\text{arrival rate of bits}}{\text{service rate of bits}}$$

"traffic intensity"



- *La/R* ~ 0: avg. queueing delay small
- *La/R* -> 1: avg. queueing delay large
- *La/R* > 1: more "work" arriving is more than can be serviced - average delay infinite!



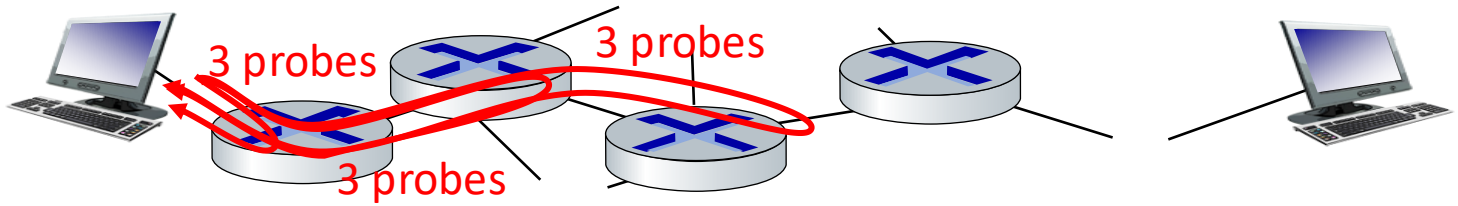
La/R ~ 0



La/R -> 1

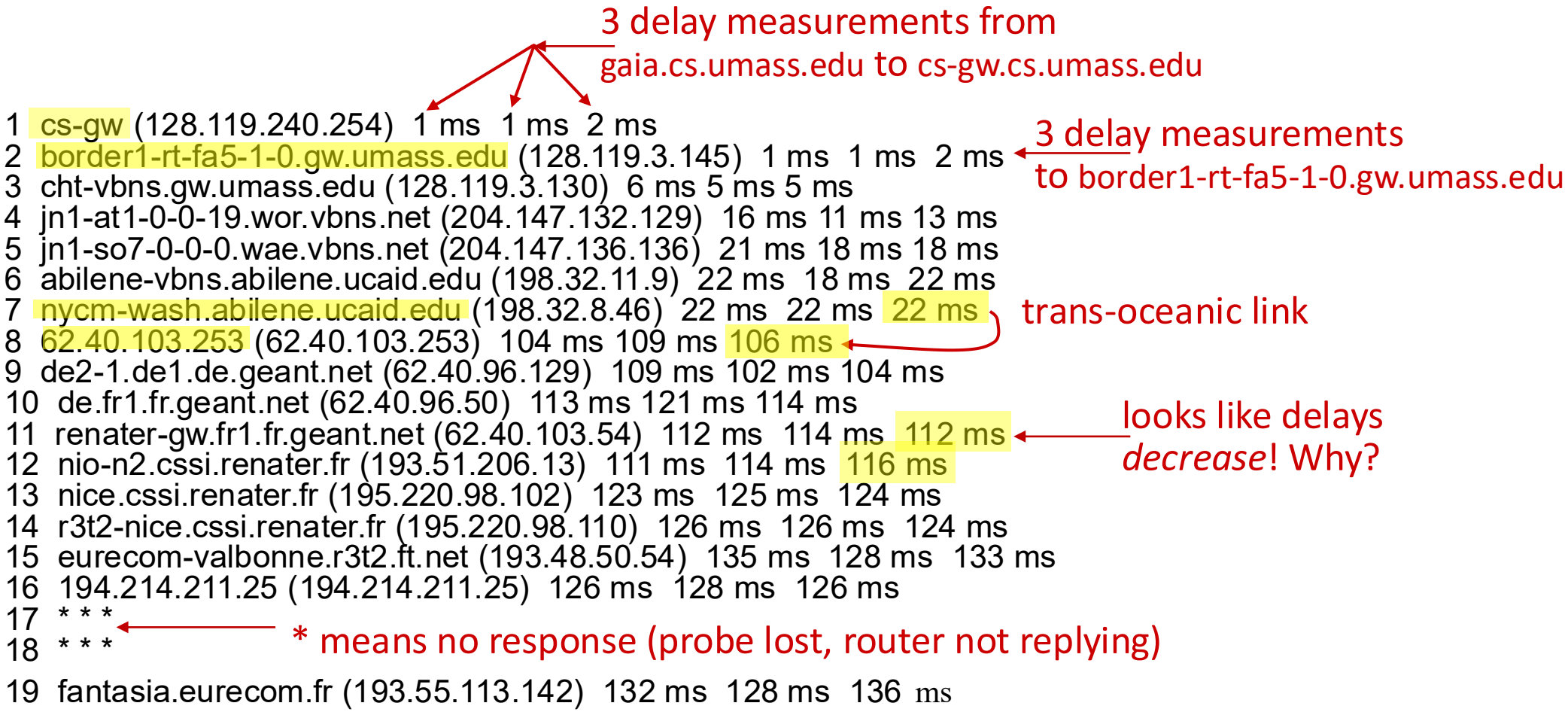
“Real” Internet delays and routes

- what do “real” Internet delay & loss look like?
- **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all *i:*
  - sends three packets that will reach router *i* on path towards destination (with time-to-live field value of *i*)
  - router *i* will return packets to sender
  - sender measures time interval between transmission and reply



Real Internet delays and routes

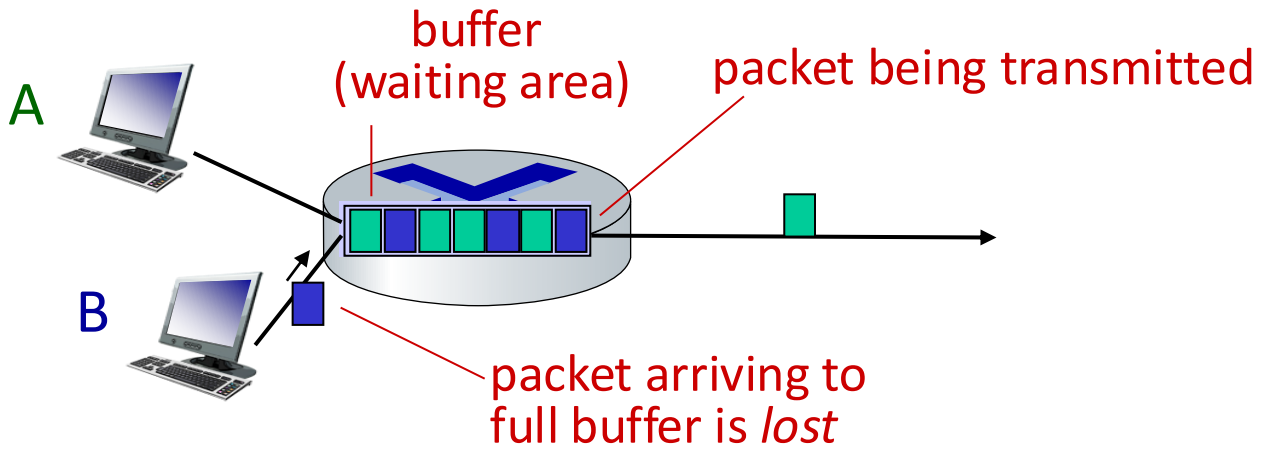
traceroute: gaia.cs.umass.edu to www.eurecom.fr



* Do some traceroutes various remote countries

Packet loss

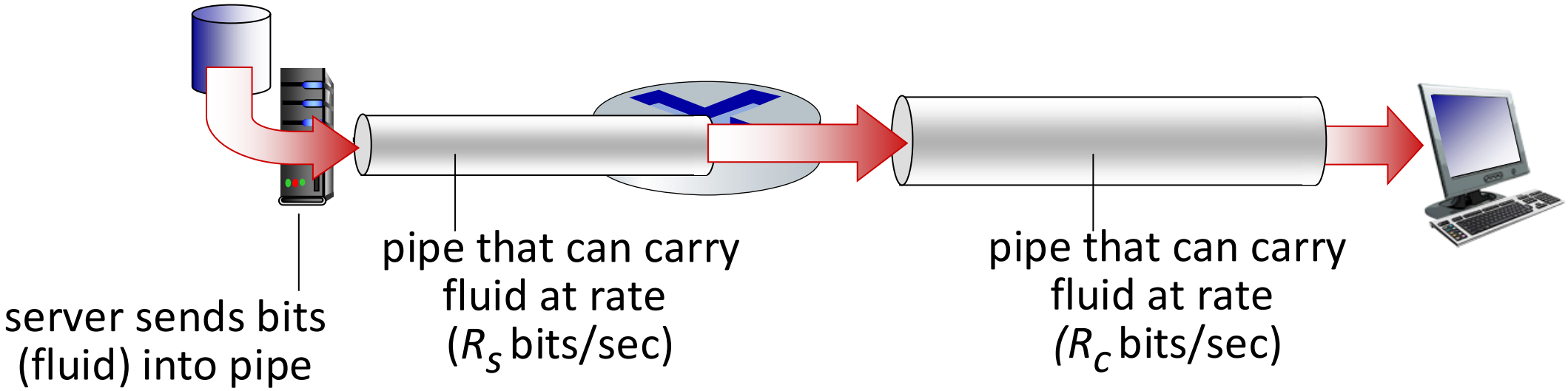
- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all



* Check out the Java applet for an interactive animation (on publisher's website) of queuing and loss

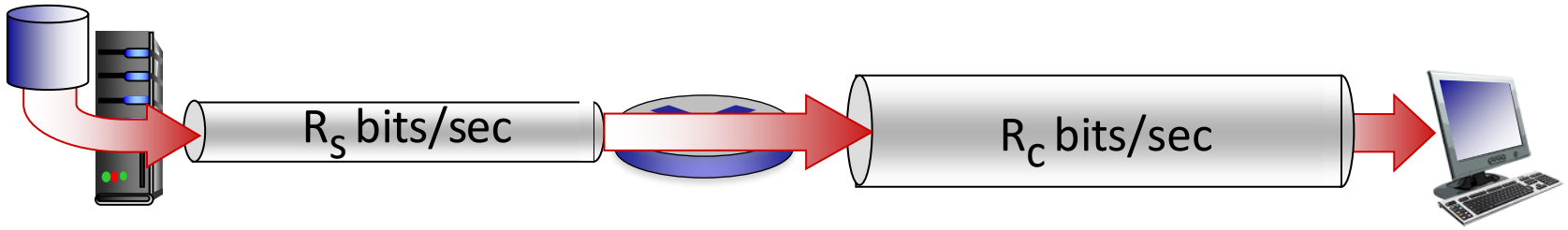
Throughput

- *throughput:* rate (bits/time unit) at which bits are being sent from sender to receiver
  - *instantaneous:* rate at given point in time
  - *average:* rate over longer period of time

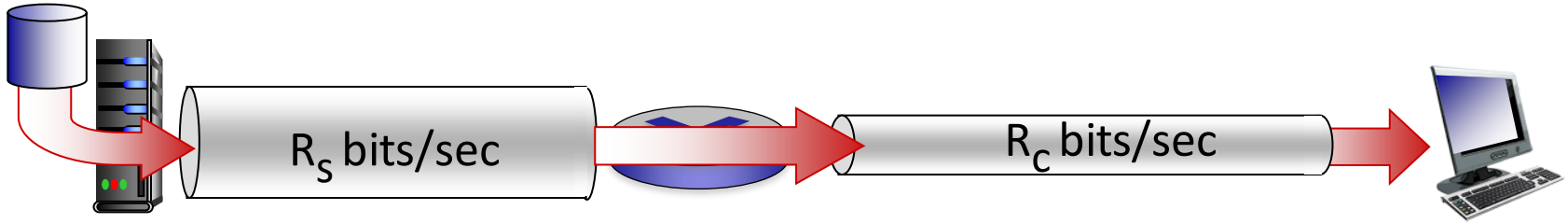


Throughput

$R_s < R_c$ What is average end-end throughput?



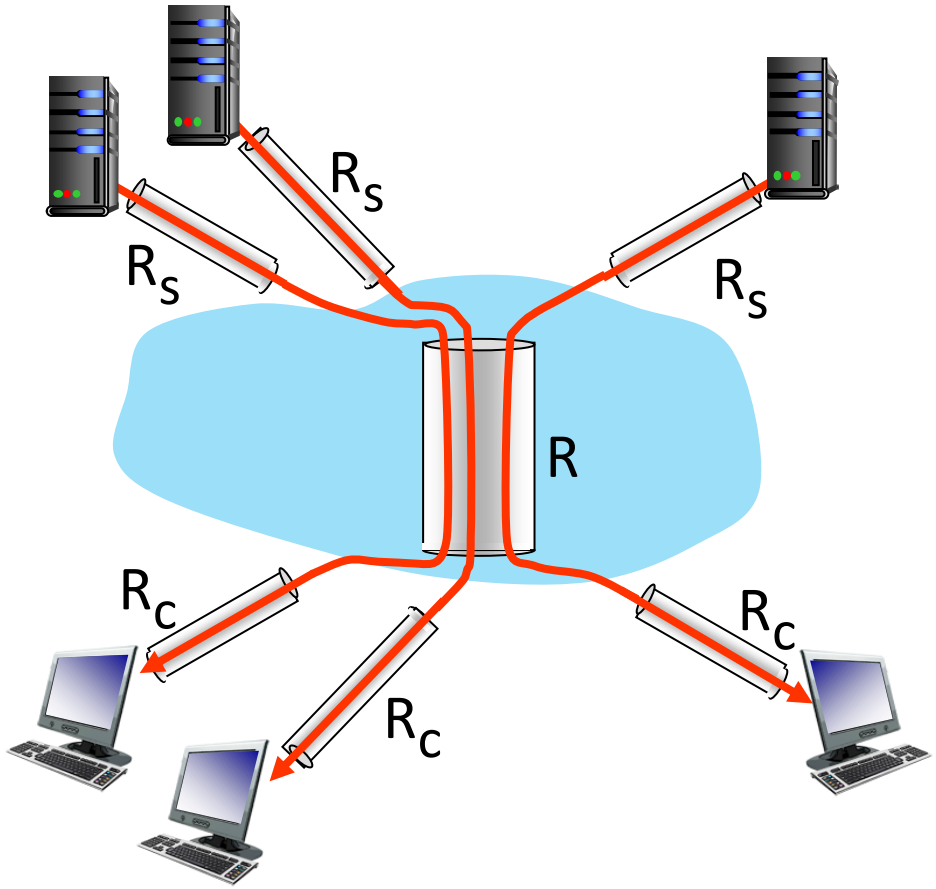
$R_s > R_c$ What is average end-end throughput?



bottleneck link

link on end-end path that constrains end-end throughput

Throughput: network scenario



3 connections (fairly) share backbone bottleneck link *R* bits/sec

- per-connection end-end throughput: $min(R_c, R_s, R/3)$
- in practice: R_c or R_s is often bottleneck

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/

summary

- Internet overview
- what's a protocol?
- network edge, access network, core
  - packet-switching versus
  - Internet structure
- performance: loss, delay, throughput
- layering, service models

You now have:

- context, overview, vocabulary, "feel" of networking
- more depth, detail, *and fun* to follow*!*